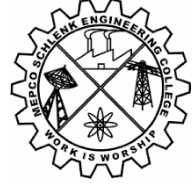




SUPERMARKET VS RETAIL ANALYSIS



A MINI PROJECT REPORT

Submitted by

P N BISWAJITH(9517202409015)

P PANDIYARAJAN(9517202409081)

B RONALD PACKIAN(9517202409094)

in partial fulfillment for the requirement of the course

23AD453 – MINI PROJECT II: DATA ANALYTICS



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
MEPCO SCHLENK ENGINEERING COLLEGE (AUTONOMOUS)
SIVAKASI**

MAY 2026

BONAFIDE CERTIFICATE



This is to certify that it is the bonafide work done by **PN.Biswajith (9517202409015), P.Pandiyarajan(9517202409081), B.RonaldPackian(951720249094)** for the course **23AD453 Mini Project – II: DATA ANALYTICS** work entitled **“SUPERMARKET VS RETAIL ANALYSIS”** at Department of Artificial Intelligence & Data Science, Mepco Schlenk Engineering College, Sivakasi during the year 2025 – 2026.

.....
Faculty In-Charge

.....
Head of the Department

Submitted for the Practical Examination held at Mepco Schlenk Engineering College,
Sivakasi on

.....
Internal Examiner - I

.....
Internal Examiner - II

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ABSTRACT

This project presents the design and development of a **Supermarket vs Retail Analysis Platform**, a comprehensive data analytics system aimed at analyzing and visualizing retail sales and customer behavior using real-world retail datasets. The study utilizes a structured retail dataset containing detailed information about products, customer transactions, store locations, sales performance, pricing, discounts, and regional trends to derive meaningful business insights.

The project begins with data preprocessing techniques such as data cleaning, handling missing values, normalization, and feature selection using tools like WEKA and Orange to ensure data quality and reliability. A data warehouse model is designed using star and snowflake schemas, enabling efficient storage and querying of large-scale retail data. OLAP operations including roll-up, drill-down, slicing, dicing, and pivoting are performed to analyze multidimensional retail data and extract useful business patterns.

Various data mining techniques are implemented to gain deeper insights. Association rule mining (Apriori and FP-Growth) is used to discover relationships between products and customer purchasing behavior, while classification algorithms such as Decision Trees, Naïve Bayes, and KNN are applied to predict sales performance and customer buying patterns. Clustering techniques including K-Means and Agglomerative clustering are utilized to group similar customer segments, product categories, and sales trends.

The system is further enhanced with an interactive dashboard built using Python and visualization libraries, providing features such as sales analysis dashboards, regional sales reports, customer behavior analysis, product performance evaluation, and seasonal trend analysis. This platform enables users to explore retail data dynamically and understand factors influencing sales growth, profitability, and customer preferences.

Overall, the project demonstrates how data warehousing, data mining, and visualization techniques can be integrated to support business intelligence, improve decision-making, and provide a deeper understanding of retail business performance and customer behavior.

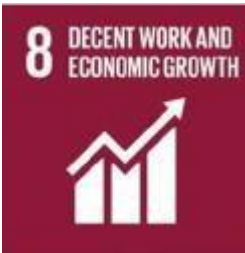
SDGs Alignment

SDG No.

Relevance of SDGs in the context of Project



The project promotes data literacy and analytical thinking, core competencies in modern education. It provides learners with hands-on experience in analytics tools and visualization, enhancing digital and analytical skills that are globally in demand.



The project uses data analytics to improve business performance and efficiency, which supports economic growth through smarter decision-making. By analyzing real-world sales, revenue, or operational data, the project encourages data-informed strategies for fair growth and productivity. Skills developed in data science, programming, and analytics prepare learners for future-ready jobs, directly contributing to the goal of decent work in the digital economy.



The project demonstrates how digital technologies (Python, R, Power BI) can be leveraged to create data-driven innovation ecosystems. It models how organizations can build analytical infrastructure to support continuous improvement and informed planning. Promotes innovation in data visualization and automation, empowering industries and governments to make evidence-based policy and operational decisions.



Through data wrangling and visualization, the project helps identify inefficiencies — such as overproduction, supply-demand imbalance, or resource wastage — enabling better planning and resource utilization. Supports sustainable production by monitoring trends, reducing waste, and optimizing operations based on analytical evidence. Encourages the responsible use of data for continuous improvement in organizational processes.

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List of Symbols and Abbreviations

Symbol / Abbreviation	Description
OLAP	Online Analytical Processing
PK	Primary Key
FK	Foreign Key
GPS	Global Positioning System
EDA	Exploratory Data Analysis
ML	Machine Learning
km	Kilometers
kmph	Kilometers per hour
m	Meters
°C	Degree Celsius
%	Percentage
dB	Decibel
hPa	Hectopascal

SYNOPSIS

Module 1: Problem Statement Identification

This module is dedicated to understanding the problem at hand and structuring data effectively for analysis. It emphasizes data warehouse schemas, which form the backbone of organizing and managing large datasets. By leveraging the Star, Snowflake, and Constellation schemas, students can structure their data efficiently. Additionally, the module covers the analytical operations of Online Analytical Processing (OLAP), which facilitates multidimensional data analysis, aiding in problem scoping and decision-making.

Module 2: Data Acquisition

This module focuses on gathering data from various sources and preparing it for analysis. Students will explore different dataset types, including structured (tabular data), semi-structured (JSON, XML), and unstructured (text, images, audio). The module ensures proper categorization of attributes into numerical, categorical, ordinal, and nominal data. Further, students will learn to identify independent (features) and dependent (target) attributes, derive new attributes when necessary, and perform feature selection to retain only the most relevant attributes for analysis. The ability to source and organize high-quality data is fundamental to ensuring meaningful insights in later stages.

Module 3: Preprocessing of Datasets

Before applying machine learning algorithms, raw data must undergo preprocessing to enhance its quality and usability. This module teaches students to clean data by handling missing values, detecting and removing outliers, and eliminating duplicate records. Using Weka, students will perform data normalization to scale features within a standard range, standardization to achieve a consistent mean and variance, and transformation techniques such as one-hot encoding and discretization. Additionally, dimensionality reduction techniques, like Principal Component Analysis (PCA), will be explored to optimize model performance while reducing computational complexity.

Module 4: Model Generation

Building predictive models is the core objective of this module. It covers different machine learning techniques, including association rule mining, regression, classification, and clustering

models. Students will work with algorithms like Apriori and FP Growth for association rule mining, various regression models (linear, multiple linear, polynomial, logistic), and classification models such as Decision Trees (ID3) and Naïve Bayes. For clustering, methods like k-means, Agglomerative Clustering, and DBSCAN will be applied. This module provides a hands-on approach to selecting and implementing suitable models for diverse data-driven problems.

Module 5: Testing of Model

Once a model is built, evaluating its performance is crucial to ensure its reliability and accuracy. This module introduces various evaluation metrics tailored for different models. For classification models, key metrics include accuracy, precision, recall, F1-score, and ROC-AUC. Regression models will be evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-Squared. Clustering models will be assessed using the Silhouette Score, Dunn Index, and Davies-Bouldin Index, while association rule mining will involve Support, Confidence, and Lift. Additionally, visualization techniques such as confusion matrices, ROC curves, scatter plots, and heatmaps will help interpret model results effectively.

Module 6: Deployment

The final step in the data science workflow is deploying the developed models for practical use. This module focuses on finalizing the trained models, integrating them into a user-friendly interface, and ensuring their seamless deployment. Students will learn how to prepare the final model, develop a functional UI for interaction, and host the model on appropriate platforms. This stage bridges the gap between theoretical learning and real-world implementation, allowing users to interact with the model in a meaningful way.

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

The **Reliance vs Retail Analysis Platform** is a data-driven analytical system developed to study, compare, and visualize the business performance of Reliance Retail and other retail market segments across different regions and time periods. The project focuses on understanding how various factors such as product categories, customer behavior, sales trends, pricing strategies, seasonal demand, and regional performance influence overall retail growth and profitability.

The system uses a large-scale retail dataset containing information about sales transactions, product details, customer demographics, store locations, purchase frequency, discounts, and revenue performance. This data is first preprocessed using techniques such as data cleaning, normalization, and feature selection to ensure accuracy and consistency. Tools like WEKA and Orange are used for preprocessing and visualization of patterns in the dataset.

A data warehouse model is designed using star and snowflake schemas, where fact and dimension tables are created to efficiently organize and manage the retail data. This structure enables advanced analytical operations using OLAP techniques such as roll-up, drill-down, slicing, and dicing to explore sales trends and customer behavior from multiple perspectives.

To extract meaningful insights, various data mining techniques are applied:

- Classification algorithms (Decision Tree, Naïve Bayes, KNN) to predict customer purchasing behavior and sales performance
- Clustering methods (K-Means, Agglomerative) to group customers and products based on buying patterns
- Association rule mining (Apriori, FP-Growth) to identify relationships between products, offers, and customer preferences

These techniques help in identifying top-selling products, customer segment, seasonal sales patterns, regional performance differences, and factors affecting retail profitability.

The project also includes an interactive visualization platform developed using Python, which provides:

- Dashboard with key performance indicators
- Regional and store-wise sales analysis
- Monthly and seasonal trend analysis
- Customer purchase behavior analysis

1.2 OBJECTIVES

To analyze retail sales patterns across different product categories, regions, customer groups, and time periods using a structured retail dataset.

- To design a data warehouse model (star and snowflake schema) for efficient storage, organization, and retrieval of large-scale retail data.
- To perform data preprocessing including data cleaning, handling missing values, normalization, and feature selection using tools like WEKA and Orange to improve data quality.
- To apply OLAP operations such as roll-up, drill-down, slicing, dicing, and pivoting for multidimensional analysis of retail sales and customer data.
- To identify key factors influencing sales performance and customer purchasing behavior, including prices.
- To implement classification algorithms (Decision Tree, Naïve Bayes, KNN) for predicting customer buying behavior and sales performance.
- To perform clustering analysis (K-Means and Agglomerative clustering) to group customers and products based on similar purchasing patterns and detect hidden business trends.
- To discover relationships between products and customer preferences using association rule mining techniques such as Apriori and FP-Growth.
- To develop an interactive visualization platform that provides dashboards, sales reports, and analytical charts for better understanding of retail business performance.

1.3 SCOPE OF THE PROJECT

The **Supermarket vs Retail Analysis Platform** focuses on analyzing and visualizing retail sales and customer data to understand business patterns, trends, and influencing factors across different regions and time periods. The scope of this project includes the collection, preprocessing, storage, analysis, and visualization of retail-related data to support data-driven business insights.

The project covers the design of a data warehouse system using star and snowflake schemas, enabling efficient organization of retail data into fact and dimension tables. It supports multidimensional analysis through OLAP operations such as roll-up, slicing, drill-down, and dicing for better exploration of sales performance and customer behavior.

The analytical scope includes the application of multiple data mining techniques such as:

- Classification (Decision Trees, Naïve Bayes, KNN) to predict customer purchasing behavior and sales performance
- Clustering (K-Means, Agglomerative) to identify customer segments and purchasing patterns
- Association rule mining (Apriori, FP-Growth) to discover relationships between products and customer preferences

These techniques help in identifying key factors affecting retail performance, such as pricing strategies, discounts, seasonal demand, product categories, and regional sales trends.

The project also includes the development of an interactive visualization system using Python, which allows users to explore retail data through dashboards, reports, and analytical charts. Users can analyze product-wise sales, customer behavior, regional trends, seasonal sales patterns, and overall retail performance in an intuitive manner.

However, the scope is limited to the available dataset and does not include real-time transaction processing or live retail data integration. The predictions and insights are based on historical retail data and may vary with changing market conditions and customer preferences.

1.4 TOOLS AND PROGRAMMING LANGUAGES USED

Programming Language Used:

Programming Language Used:

- **Python:**

Used as the primary programming language for data analysis, data mining, and building the interactive retail analysis and visualization platform. It supports libraries for machine learning, visualization, and dashboard development.

Data Analysis & Visualization Tools:

- **Jupyter Notebook:**

Used for writing and executing Python code, performing exploratory data analysis, and implementing machine learning algorithms such as Decision Trees, KNN, and Naïve Bayes.

- **Plotly & Dash (Python Libraries):**

Used to develop the interactive dashboard and visualization platform, including graphs, sales reports, charts, and analytical dashboards for retail data.

Data Preprocessing Tools:

- **WEKA:**

Used for preprocessing tasks such as handling missing values, discretization, feature selection, and visualization techniques like histograms and scatter plots.

Data Mining & Machine Learning Libraries:

- **Scikit-learn (Python Library):**

Used to implement machine learning algorithms such as:

- Decision Tree (ID3, CART, C4.5)
- Naïve Bayes
- K-Nearest Neighbors (KNN)
- K-Means Clustering
- Agglomerative Clustering

- **Pandas & NumPy:**

Used for data manipulation, data cleaning, preprocessing, and numerical computations on retail datasets.

CHAPTER 2

PROBLEM STATEMENT IDENTIFICATION

2.1 UNDERSTANDING THE PROBLEM

Retail business operations generate large volumes of data from sales transactions, customer purchases, product inventories, and regional market activities. This data is influenced by multiple factors such as customer preferences, pricing strategies, seasonal demand, product categories, discounts, and geographic regions. However, retail data is often high-dimensional and difficult to analyze effectively using traditional methods.

The primary problem addressed in this project is the lack of an efficient system to analyze and extract meaningful insights from large-scale retail data. The dataset contains numerous attributes such as product categories, customer details, sales records, store locations, discounts, and transaction history, making it challenging to identify patterns, relationships, and trends without proper analytical techniques.

To better understand the problem, a set of analytical questions is defined, focusing on key aspects such as:

- Identifying frequently purchased products and high-performing product categories
- Determining factors that influence sales performance and customer purchasing behavior
- Analyzing the impact of pricing, discounts, and seasonal demand on retail sales
- Identifying top-performing regions, stores, and customer segments
- **These questions highlight the need to:**
- Handle large and multidimensional retail datasets efficiently
- Identify relevant features affecting customer behavior and sales performance
- Apply analytical and machine learning techniques to uncover hidden business patterns

2.2 DATA WAREHOUSE DESIGN

2.2.1 STAR SCHEMA

Star schema is a dimensional design for a relational database often used in a data warehouse system. The star schema has a fact table at the center, and some dimension tables surround the fact table. The star schema name comes from the appearance of the schema, which looks like a star. In the star schema, related dimensions are grouped as columns in dimension tables and used to store the context of the facts stored in the fact table.

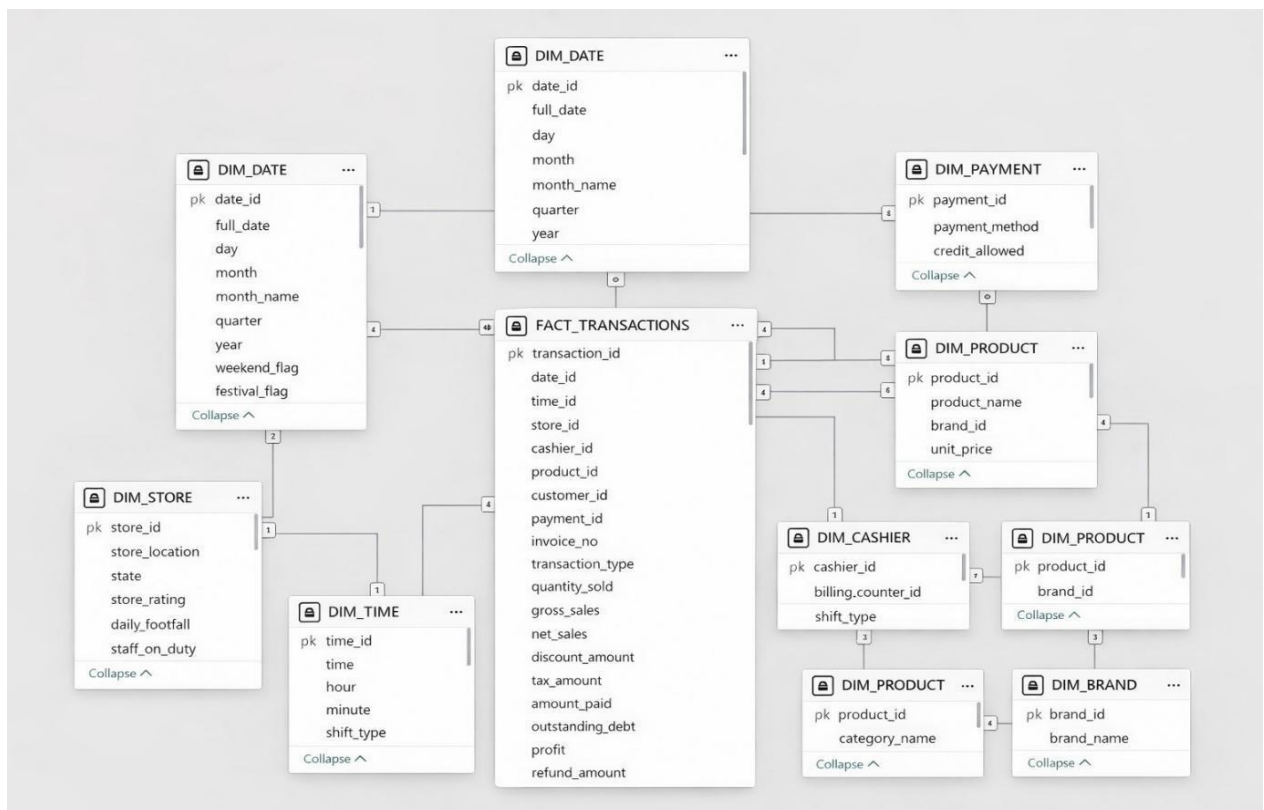


Fig no 2.2.1.Star Schema

Star schema description:

The Retail Analysis System follows a Star Schema design in which the central fact table stores transaction-related data and is connected to multiple dimension tables for detailed analytical processing.

Fact Table

FACT_TRANSACTIONS is the fact table containing the attributes:

(Transaction_ID, Date_ID, Time_ID, Store_ID, Cashier_ID, Product_ID, Customer_ID, Payment_ID, Invoice_No, Transaction_Type, Quantity_Sold, Gross_Sales, Net_Sales, Discount_Amount, Tax_Amount, Amount_Paid, Outstanding_Debt, Profit, Refund_Amount)

This table references all the dimension tables and stores measurable retail transaction data usebusiness analysis.

Dimension

Tables

Date

Dimension

Table

DIM_DATE contains the attributes:

(Date_ID, Full_Date, Day, Month, Month_Name, Quarter, Year, Weekend_Flag, Festival_Flag) This table helps in analyzing sales trends based on daily, monthly, quarterly, and yearly periods.

Time Dimension Table

DIM_TIME contains the attributes:

(Time_ID, Time, Hour, Minute, Shift_Type)

This table is used to analyze sales performance during different time periods and work shifts.

Store Dimension Table

DIM_STORE contains the attributes:

(Store_ID, Store_Location, State, Store_Rating, Daily_Footfall, Staff_On_Duty)

This table stores information related to store branches and regional business performance.

Product Dimension Table

DIM_PRODUCT contains the attributes:

(Product_ID, Product_Name, Brand_ID, Unit_Price, Category_Name)

DIM_BRAND contains the attributes:

(Brand_ID, Brand_Name)

This table stores information about product brands used for brand-wise sales analysis.

Cashier Dimension Table

DIM_CASHIER contains the attributes:

(Cashier_ID, Billing_Counter_ID, Shift_Type)

This table is used to analyze cashier performance and billing activities.

Payment Dimension Table

DIM_PAYMENT contains the attributes:

(Payment_ID, Payment_Method, Credit_Allowed)

This table stores payment-related information such as payment modes and credit facilities.

2.2.2 SNOWFLAKE SCHEMA

Snowflake schema consists of a fact table surrounded by multiple dimension tables which can be connected to other dimension tables via many-to-one relationship. The snowflake schema is a kind of star schema however it is more complex than a star schema in terms of the data model. This schema resembles a snowflake, therefore, it is called the snowflake schema. A snowflake schema is designed from the star schema by further normalizing dimension tables to eliminate data redundancy. Therefore, in the snowflake schema, instead of having big dimension tables connected to a fact table, we have a group of multiple dimension tables. In the snowflake schema, dimension tables are normally in the third normal form (3NF). The snowflake schema helps save storage however it increases the number of dimension tables.

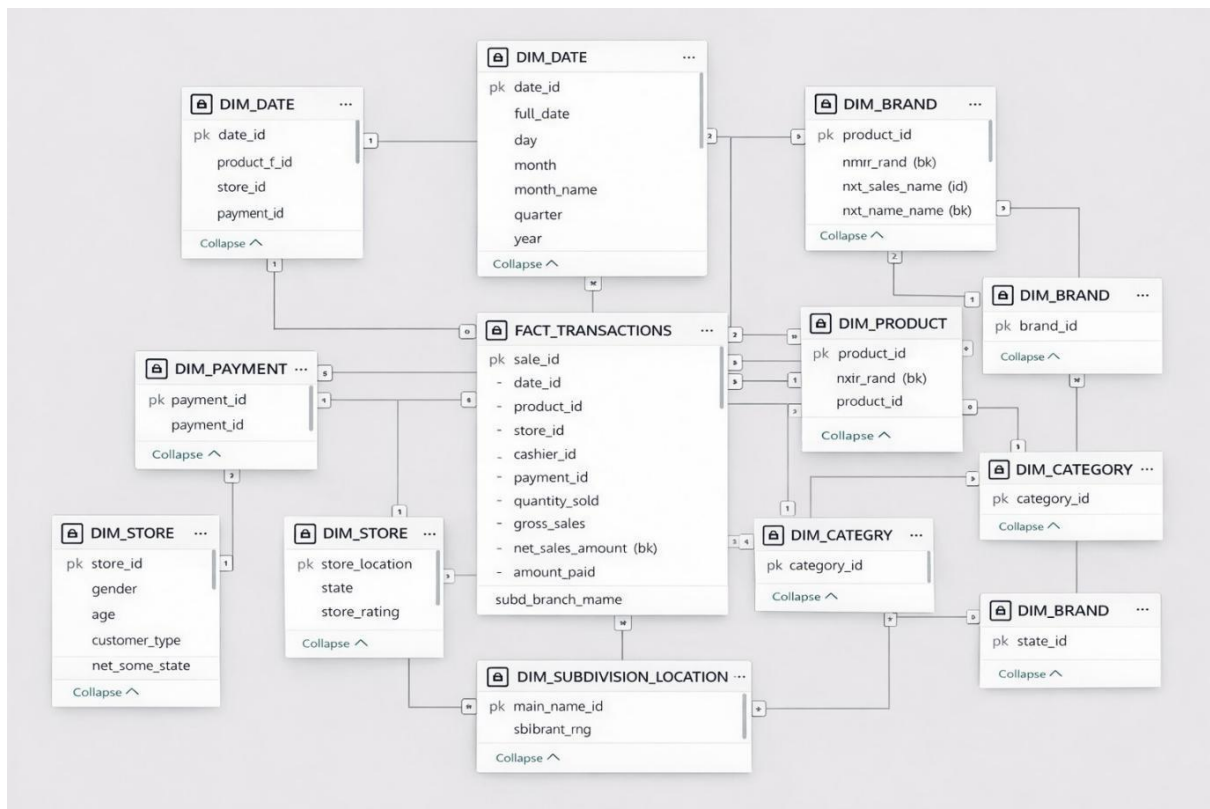


Fig no 2.2.2.Snowflake Schema

Snowflake Schema Description:

The Retail Analysis System follows a Snowflake Schema design in which the fact table is connect multiple normalized dimension tables. This structure reduces data redundancy and improves data organization for advanced analytical processing.

FACT_TRANSACTIONS

The FACT_TRANSACTIONS table contains the following attributes:

Sale_ID (PK), Date_ID (FK), Product_ID (FK), Store_ID (FK), Cashier_ID (FK)
Quantity_Sold, Gross_Sales, Net_Sales_Amount, Amount_Paid, Sub_Branch_Name

This table stores transactional retail data and references all related dimension tables.

Dimen

sion

Tables

DIM_

DATE

Date_ID (PK), Product_F_ID, Store_ID, Payment_ID

This table stores date-related details used for time-based sales analysis.

DIM_CALENDAR

Date_ID (PK), Full_Date, Day, Month, Month_Name, Quarter, Year

This table provides detailed calendar information for analyzing monthly, quarterly, yearly trends.

DIM_PRODUCT

Product_ID (PK), Brand_ID (FK), Category_ID (FK)

This table stores product-related information and connects with brand and category tables.

DIM_BRAND

Brand_ID (PK), Brand_Name

This table contains information about different product brands used for brand-wise analysis.

DIM_CATEGORY

Category_ID (PK), Category_Name

This table stores product category details for category-based sales analysis.

DIM_STORE

Store_ID (PK), Store_Location, State, Store_Rating

DIM_CUSTOMER

Customer_ID (PK), Gender, Age, Customer_Type, Net_Some_State

This table stores customer demographic information used for customer behavior analysis.

DIM_PAYMENT

Payment_ID (PK), Payment_Method

This table stores payment-related details such as payment modes used by customers.

DIM_SUBDIVISION_LOCATION

Main_Branch_ID (PK), Sub_Branch_Name

This table contains branch and subdivision details for branch-level performance analysis.

DIM_CASHIER

Cashier_ID (PK), Billing_Counter_ID, Shift_Type

This table stores cashier-related information for billing and staff performance analysis.

OLAP AND ANALYTICAL QUESTIONS

Roll up:

The roll-up operation (also called drill-up or aggregation operation) performs aggregation on a data cube, either by climbing up a concept hierarchy for a dimension or by climbing down a concept hierarchy, i.e. dimension reduction.

Row Labels	Sum of net_sales
Aashirvaad	14711203.06
Britannia	14711203.06
Bru	14711203.06
Cadbury	14711203.06
Colgate	14711203.06
Fortune	14711203.06
Head & Shoulders	14711203.06
India Gate	14711203.06
Lays	14711203.06
Madhur	14711203.06
Maggi	14711203.06
Pears	14711203.06
Surf Excel	14711203.06
Tata Salt	14711203.06
Tata Tea	14711203.06
Vim	14711203.06
Grand Total	14711203.06

Fig no 2.3.1

Drill down:

The drill down operation (also called roll down) is the reverse of roll up. It navigates from less detailed data to more detailed data. It can be realized by either stepping down a concept hierarchy for a dimension or introducing additional dimensions. Performing roll down operation on the same cube mentioned above:

Product_Category	Product	Sum of net_sales	Sum of tax_percentage	Sum of quantity_sold	Sum of discount_percentage
→ Food		1,4711,203.06	65,831	28,972	83,482.76
▼ Food					
	Basmati Rice Premium	1,4711,203.06	65,831	28,972	83,482.76
	Digestive Biscuits	1,4711,203.06	65,831	28,972	83,482.76
	Instant Coffee	1,4711,203.06	65,831	28,972	83,482.76
	Iodized Cooking Salt	1,4711,203.06	65,831	28,972	83,482.76
	Milk Chocolate Bar	1,4711,203.06	65,831	28,972	83,482.76
	Potato Chips Salted	1,4711,203.06	65,831	28,972	83,482.76
	Sunflower Oil 1L	1,4711,203.06	65,831	28,972	83,482.76
	Tea Powder Strong	1,4711,203.06	65,831	28,972	83,482.76
→ Household		1,4711,203.06	18,456	7,672	28,094.00
	Bath Soap Herbal		65,831	7,672	28,094.00
	Dishwash Liquid		18,456	7,672	28,094.00
	Instant Noodles Pack		18,456	7,672	28,094.00
	Laundry Detergent		18,456	7,672	28,094.00
	Refined Sugar		18,456	7,672	28,094.00
	Shampoo Anti-Dandruff		18,456	7,672	28,094.00
	Toothpaste Gel		18,456	7,672	28,094.00
	Wheat Flour Select		18,456	7,672	28,094.00
Grand Total		1,4711,203.06	84,287	36,644	111,576.76

Fig no 2.3.3 Drill down

Slicing:

Slicing is an OLAP operation used to extract a specific subset of data by fixing a single dimension to a particular value. It reduces a multi-dimensional data cube into a smaller view, making analysis more focused and easier. Slicing helps in studying specific patterns.

Column Labels					
FALSE		TRUE		Total Sum of Discount_Percentage	Total Sum of MRP
Row Labels	Sum of Discount_Percentage	Sum of MRP	Sum of Discount_Percentage	Sum of MRP	
(blank)	53.94	1292.45	77.76	902.54	131.7
Grand Total	53.94	1292.45	77.76	902.54	131.7

Fig no 2.3.5 Slicing

Dicing:

Dicing is an OLAP operation used to extract a smaller sub-cube from a data cube by applying multiple conditions on one or more dimensions at the same time. Unlike slicing, which fixes a single dimension to one value, dicing selects a range or set of values across dimensions.

Row Labels	Sum of gross_sales	Count of company_name	Sum of gross_sales	Count of company_name	Sum of gross_sales	Count of company_name	Sum of gross_sale
City	71815.92	27	71815.92	27	71815.92	27	71815.9
Town	129320.12	27	129320.12	27	129320.12	27	129320.1
Village	99264.99	27	99264.99	27	99264.99	27	99264.9
Basmati Rice Premium							
City	393986.48	290	393986.48	290	393986.48	290	393986.4
Town	437496.44	290	437496.44	290	437496.44	290	437496.4
Village	186543.3	290	186543.3	290	186543.3	290	186543.3
Bath Soap Herbal							
City	426958.61	297	426958.61	297	426958.61	297	426958.6
Town	397142.86	297	397142.86	297	397142.86	297	397142.8
Village	203589.52	297	203589.52	297	203589.52	297	203589.5
Digestive Biscuits							
City	474099.79	331	474099.79	331	474099.79	331	474099.7
Town	429443.91	331	429443.91	331	429443.91	331	429443.9
Village	216000.7	331	216000.7	331	216000.7	331	216000.7
Dishwash Liquid							
City	392119.55	284	392119.55	284	392119.55	284	392119.5
Town	428742.53	284	428742.53	284	428742.53	284	428742.5
Village	187062.6	284	187062.6	284	187062.6	284	187062.6
Instant Coffee							

Fig no 2.3.7 Dicing

Pivot:

Pivot otherwise known as Rotate changes the dimensional orientation of the cube, i.e. It rotates the data axes to view the data from different perspectives. Pivot groups data with different dimensions.

Sum of Net_Sales_Value	Column Labels	
Row Labels	(blank)	Grand Total
Jan	397.50318	397.50318
Mar	7104.227508	7104.227508
Apr	2596.794717	2596.794717
Sep	358.201753	358.201753
Dec	1618.71336	1618.71336
Grand Total	12075.44052	12075.44052

Fig no 2.3.9 Pivot

Sum of Store_Footfall	Column Labels			
Row Labels	Evening	Morning	Night	Grand Total
(blank)	1142	1627	6014	8783
Grand Total	1142	1627	6014	8783

Fig no 2.3.10 Pivot

Dimension tables and Its Pivot Tables:

Column Labels v		
(blank)		Total Count of Net_Sales Total Sum of Unit_Price
Row Labels v	Count of Net_Sales	Sum of Unit_Price
Jan	290.7	290.7
Mar	673.25	673.25
Apr	594.32	594.32
Sep	366.05	366.05
Dec	272.8	272.8
Grand Total	2197.12	2197.12

Fig no 2.3.11

Sum of Quantity_Sold		Column Labels v							Grand Total
Row Labels v		1.16	1.41	6.79	7.95	36.63	37.53	39.07	
= (blank)		10	12	1	1	2	11	9	46
= 2023		1	12		1	2	11		27
Qtr1			12			2			14
Qtr2		1					11		12
Qtr3					1				1
= 2024		9		1				9	19
Qtr1		9							9
Qtr3				1					1
Qtr4								9	9
Grand Total		10	12	1	1	2	11	9	46

Fig no 2.3.12

CHAPTER 3

DATAACQUISITION

3.1 DATA COLLECTION METHODS

The dataset used in this project is obtained from an online open-source platform, Kaggle, which provides real-world datasets for research and analysis. The dataset titled Retail Sales Dataset contains structured records of retail transactions collected from different stores, regions, and customer groups.

The data collection is based on:

- Sales transactions recorded from retail stores and billing systems
- Customer purchase history and product sales records
- Store management systems capturing product, pricing, and inventory details
- Market and customer behavior data including discounts, seasonal demand, and payment methods

The dataset is downloaded in Excel/CSV format and imported into the system for further analysis. It consists of multiple attributes related to products, customers, store locations, payment performance, and transaction details , making it suitable for multidimensional retail analysis.

Dataset Description:

Source: Kaggle

Link: [Retail Sales Dataset on Kaggle](#)

Number of columns: 40+

Number of rows: 10000+

Size: Approximately 3–5 MB

TYPES OF DATA :

Column Name	Data Type
Transaction_ID	Categorical
Customer_ID	Categorical
Customer_Name	Categorical
Gender	Categorical
Age	Numerical
Product_ID	Categorical
Product_Name	Categorical
Category_Name	Categorical
Brand_Name	Categorical
Store_ID	Categorical
Store_Location	Categorical
Region	Categorical
Payment_Method	Categorical
Invoice_No	Categorical
Quantity_Sold	Numerical
Unit_Price	Numerical
Discount_Amount	Numerical
Tax_Amount	Numerical
Gross_Sales	Numerical
Net_Sales	Numerical
Profit	Numerical
Amount_Paid	Numerical
Outstanding_Debt	Numerical
Transaction_Date	Categorical
Month	Categorical
Year	Numerical
Seasonal_Demand	Categorical
Festival_Flag	Categorical
Customer_Type	Categorical
Customer_Feedback	Categorical
Store_Rating	Numerical
Daily_Footfall	Numerical
Billing_Counter_ID	Categorical
Shift_Type	Categorical
Staff_On_Duty	Numerical
Product_Returned	Categorical
Refund_Amount	Numerical
Sales_Performance	Categorical
Sales_Channel	Categorical
Delivery_Status	Categorical
Inventory_Level	Numerical
Observation_Counts	Numerical
Sales_Trend	Categorical

Table no 3.2.1 Types of data

3.2 ATTRIBUTE CATEGORIZATION VS DEPENDENT ATTRIBUTES

3.2.1 INDEPENDENT VS DEPENDENT ATTRIBUTES

Independent Attributes (Features):

These are the variables that influence or explain the outcome. They are used as input models.

Dependent Attributes (Target Variables):

These are the outcomes that the system aims to predict, analyze, or understand.

1. Which products are most frequently purchased by customers, and what common factors explain these preferences?

Target: Product_Name, Category_Name

Supporting: Customer_Type, Region, Brand_Name, Payment_Method, Seasonal_Demand

2. What are the primary reasons behind high sales performance across different product categories? Target: Net_Sales

Supporting: Product_Category, Discounts, Customer_Type, Store_Rating

3. What factors most often lead to poor sales performance in retail stores? Target: Net_Sales (Low Sales Cases)

Supporting: Discounts, Product_Category, Region, Customer_Feedback

4. How does product pricing influence retail sales performance? Target: Net_Sales

Supporting: Unit_Price, Discount_Amount, Product_Category, Quantity_Sold

5. Which product categories show the most consistent sales trends over time? Target: Sales_Trend

Supporting: Product_Category, Year, Region, Quantity_Sold

6. Which regions act as major sales hubs for multiple product categories? Target: Region

Supporting: Product_Category, Gross_Sales, Quantity_Sold

7. Which sales channels experience the highest profit rates? Target: Profit

Supporting: Sales_Channel, Product_Category, Payment_Method

8. Which store locations are most prone to low sales performance?
Target: Net_Sales
Supporting: Store_Location, Product_Category, Customer_Type
9. How do geographic characteristics influence retail sales outcomes?
Target: Sales_Performance
Supporting: Region, Store_Location, Customer_Type, Product_Category
10. Which store locations function as both major sales and distribution centers?
Target: Store_Location
Supporting: Observation_Counts, Product_Category
11. How do sales performance rates vary across different product categories?
Target: Net_Sales
Supporting: Product_Category, Seasonal_Demand
12. Which seasons show peak retail sales activity and why?
Target: Observation_Counts
Supporting: Seasonal_Demand, Festival_Flag, Product_Category
13. How have retail sales patterns changed over the years?
Target: Gross_Sales
Supporting: Product_Category, Region, Year
14. How do sales failures occur most across different market conditions?
Target: Net_Sales
Supporting: Discounts, Customer_Feedback, Seasonal_Demand
15. Which product categories show the highest sales frequency and customer demand?
Target: Quantity_Sold, Observation_Counts
Supporting: Product_Category
16. Which product categories demonstrate the highest adaptability in changing market conditions?
Target: Net_Sales, Sales_Trend
Supporting: Product_Category, Customer_Type, Seasonal_Demand
17. How do market conditions impact retail sales performance across categories?

Target: Net_Sales

Supporting: Seasonal_Demand, Discounts, Product_Category

18. What patterns distinguish low-selling products from

high-selling products? Target: Quantity_Sold

Supporting: Net_Sales, Product_Category, Customer_Type

19. What common characteristics are observed in

profitable retail stores? Target: Profit

Supporting: Product_Category, Customer_Type, Payment_Method

20. Which product categories face the greatest sales challenges, and what factors contribute to these difficulties?

Target: Net_Sales

Supporting: Product_Category, Discounts, Seasonal_Demand, Customer_Feedback

3.2.2 FEATURE SELECTION AND EXTRACTIONS

Feature Selection

Feature selection is the process of identifying and selecting the most relevant attributes from the dataset that contribute significantly to the target variable. Since the retail dataset contains a large number of attributes such as product categories, customer details, sales records, payment methods, store locations, discounts, and transaction details, not features are equally important for every analysis.

- To predict sales performance, features like Product_Category, Discount_Amount, Customer_Type,

Seasonal_Demand, and

Payment_Method are selected. By

selecting only relevant features:

- The complexity of the model is reduced
- Processing time is minimized
- Accuracy of classification and clustering algorithms improves

Feature Extraction

Feature Extraction is the process of transforming raw retail data a more meaningful and suitable format for analysis. It involves creating new features or modifying existing ones to improve the efficiency of machine learning models.

In the retail dataset, feature extraction is applied in several ways:

Encoding categorical data:

Attributes such as Product_Category, Payment_Method, Customer_Type, and Store_Location are converted into numerical form using techniques like label encoding and one-hot encoding.

Discretization:

Continuous attributes such as Gross_Sales, Net_Sales, and Quantity_Sold are converted into categories like low, medium, and high sales, making analysis easier.

Data transformation:

Numerical values are normalized or scaled to maintain uniformity across features and improve analytical performance.

CHAPTER 4

DATA PREPROCESSING

4.1 Data Cleaning

4.1.1 Handling Missing values

Using Orange

If dataset contains missing values, handle them using the "Impute" widget. This widget provides options for imputing missing values using various techniques such as mean, median, or most frequent value imputation.

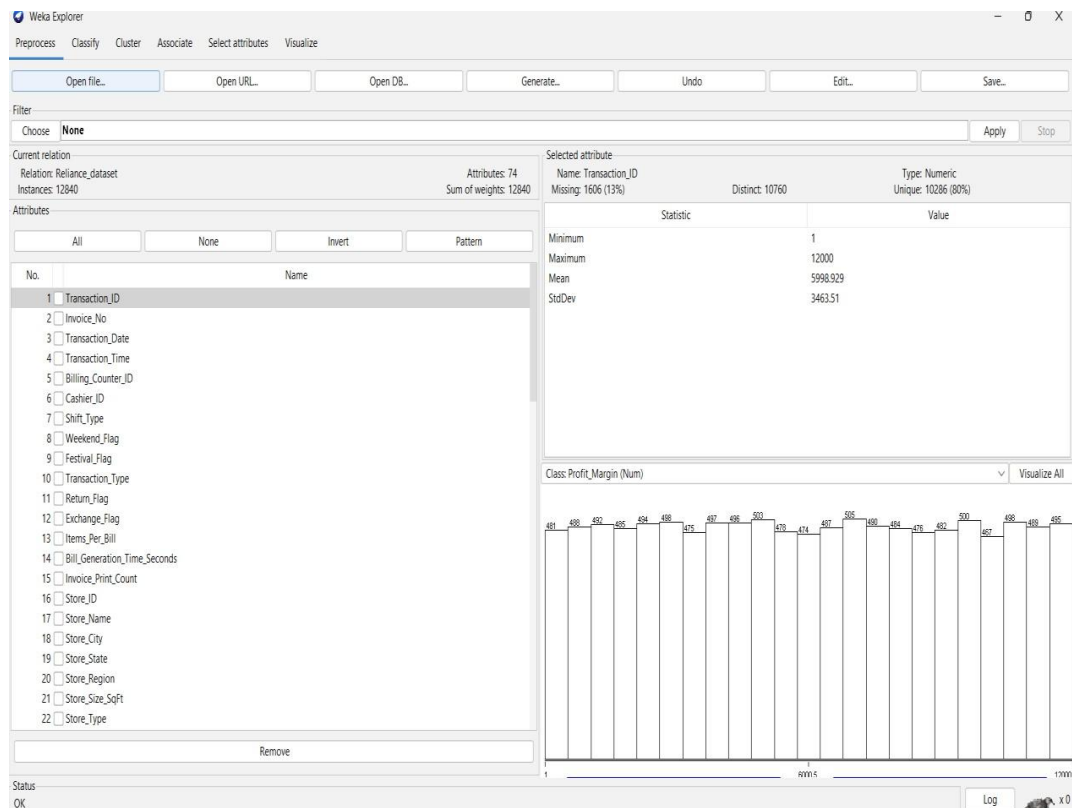


Fig 4.1.1 Dataset with missing values

Handle Missing Values using WEKA

Missing values can adversely affect the performance of machine learning models. Weka offers several options for handling missing values:

Use the "ReplaceMissingValues" filter to replace missing values with the mean, median, mode, or a user-specified value.

Another option is to use more advanced techniques like imputation, which can be implemented using filters such as "SimpleImputer" or "IterativeImputer."

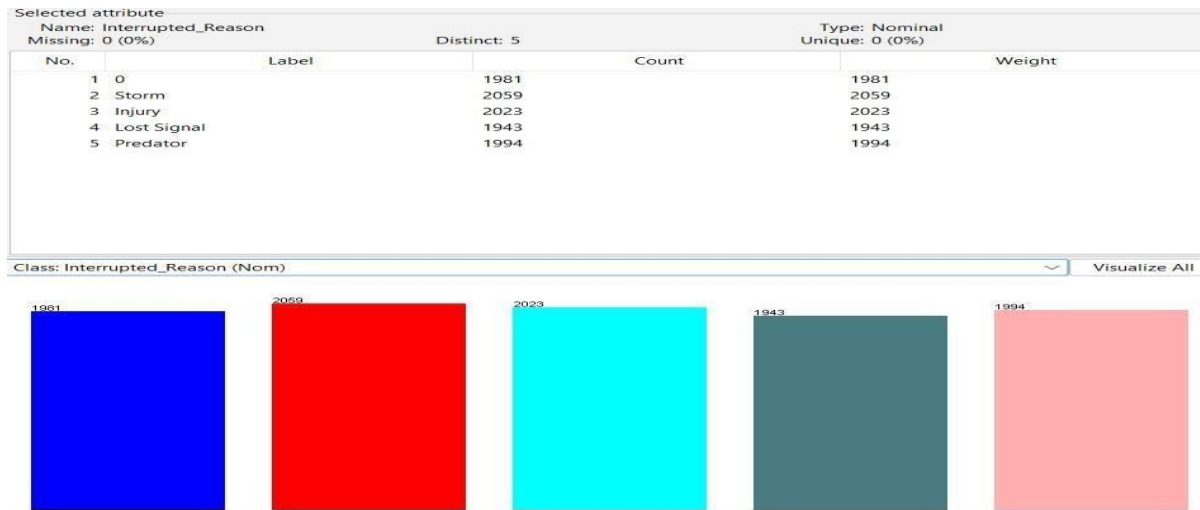


Fig 4.1.3 Before Handling missing values

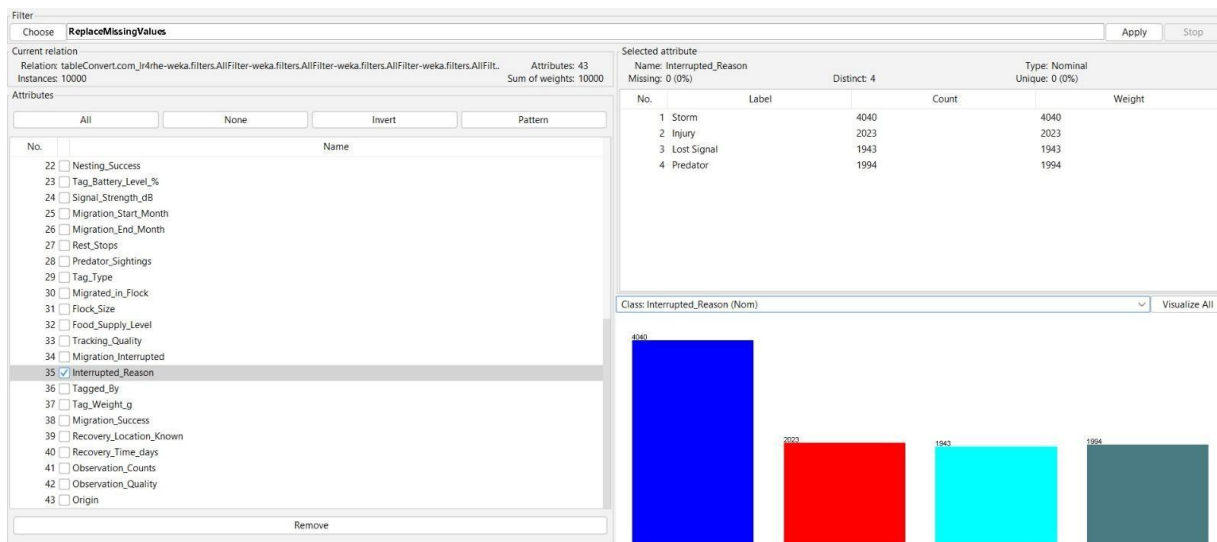


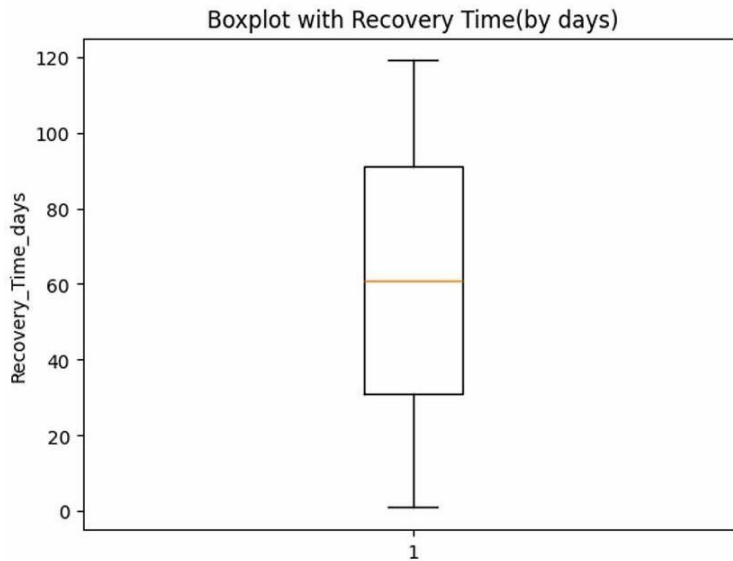
Fig 4.1.4 After handling Missing values

Outlier detection

```
plt.boxplot(bird["Recovery_Time_days"])

# Add title and Labels
plt.title("Boxplot with Recovery Time(by days)")
plt.ylabel("Recovery_Time_days")

# Display the plot
plt.show()
```



Outlier analysis was performed using box plots to identify any extreme values in the dataset. After visualizing the numerical attributes such as flight distance, duration, speed, and environmental factors, it was observed that no significant outliers were present.

The values were found to be within the acceptable range, indicating that the dataset is clean and consistent

4.2 Data Transformation

Data discretization using WEKA

Data discretization is the process of converting continuous numerical values into categorical values (bins or intervals).

Fig 4.1.5 Before Discretization

Fig 4.1.6 After Discretization

Transaction_ID	Customer_ID	Transaction_ID	Customer_ID
nan	nan	nan	nan
2.0	16553.0	(-6.996, 1801.2]	(9961.005, 18000.0]
3.0	15759.0	(-6.996, 1801.2]	(9961.005, 18000.0]
4.0	17755.0	(-6.996, 1801.2]	(9961.005, 18000.0]
5.0	23384.0	(-6.996, 1801.2]	(18000.0, 25999.0]
6.0	39928.0	(-6.996, 1801.2]	(33998.0, 41997.0]
7.0	37478.0	(-6.996, 1801.2]	(33998.0, 41997.0]
8.0	43897.0	(-6.996, 1801.2]	(41997.0, 49996.0]
nan	nan	nan	nan
10.0	42374.0	(-6.996, 1801.2]	(41997.0, 49996.0]
11.0	30529.0	(-6.996, 1801.2]	(25999.0, 33998.0]
12.0	48409.0	(-6.996, 1801.2]	(41997.0, 49996.0]
13.0	42368.0	(-6.996, 1801.2]	(41997.0, 49996.0]
14.0	24371.0	(-6.996, 1801.2]	(18000.0, 25999.0]
nan	nan	nan	nan
16.0	13895.0	(-6.996, 1801.2]	(9961.005, 18000.0]
17.0	24453.0	(-6.996, 1801.2]	(18000.0, 25999.0]
nan	nan	nan	nan

4.2.1 Encoding Categorical Data

One hot Encoding:

One-Hot Encoding is a data preprocessing technique used to convert categorical values into a numerical format by creating separate binary (0 or 1) columns for each category.

Fig 4.2.1 – Before One-Hot Encoding (Retail Dataset)

Retail dataset - Selected categorical attributes before encoding:							
	Product_Name	Category_Name	Customer_Type	Region	Brand_Name	Payment_Method	Seasonal_Demand
0	Smartphone X1	Electronics	Regular	North	TechPro	Credit Card	High
1	Wireless Earbuds	Electronics	New	East	SoundMax	UPI	Medium
2	Running Shoes	Footwear	Regular	West	Sprint	Debit Card	High
3	Coffee Maker	Appliances	Premium	South	HomeEase	Credit Card	Low
4	Denim Jeans	Apparel	Regular	North	StyleHub	UPI	Medium
5	Laptop Pro 15	Electronics	Premium	East	TechPro	Net Banking	High
6	Smart Watch	Wearables	Regular	South	TimeTech	Credit Card	Medium
7	T-Shirt (Men)	Apparel	New	West	StyleHub	Cash on Delivery	Low
8	Blender	Appliances	Regular	Central	HomeEase	Debit Card	Medium
9	Office Chair	Furniture	Bulk	North	ComfySeat	Net Banking	High

```
from sklearn.preprocessing import OneHotEncoder

# Apply One-Hot Encoding to 'Category_Name'
encoder = OneHotEncoder(sparse_output=False,
                        handle_unknown='ignore')

encoded_data = encoder.fit_transform(
    retail_df[['Category_Name']]
)

print("\nOne-Hot Encoded Category_Name:\n",
      encoded_data)
```

This figure shows the dataset before applying One-Hot Encoding to categorical variables.

Fig 4.2.1 Before OneHot Encoding

One-Hot Encoded Data:

```
[[0. 0. 1. 0. 0. 0.]
 [0. 0. 1. 0. 0. 0.]
 [0. 0. 0. 1. 0. 0.]
 ...
 [0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 1. 0. 0.]
 [0. 0. 0. 1. 0. 0.]]
```

Fig 4.2.2 After OneHot Encoding

4.2.2 Feature Selection Techniques

Feature selection using WEKA

After adding the "AttributeSelection" filter to the pipeline, click on it to configure its parameters. You can specify which attributes you want to select or deselect by selecting them from the list of attributes.

Weka offers various feature selection algorithms, such as BestFirst, GreedyStepwise, or Ranker, CfsSubsetEval, InfoGainAttributeEval, and ReliefFAttributeEval. Choose the algorithm that best suits your data and objectives. Adjust any additional parameters specific to the chosen feature selection algorithm, such as the number of selected features or evaluation metrics.

1. Which products are most frequently purchased by customers, and what common factors explain these preferences?

Target: Product_Name, Category_Name

Supporting: Customer_Type, Region, Brand_Name, Payment_Method, Seasonal_Demand

Fig no. 4.1.10 After applying filter for Invoice_No

The screenshot displays the WEKA software interface for attribute selection. The 'Current relation' panel shows the dataset 'Retail_Analytics_Dataset' with 49 attributes and 9000 instances. The 'Attributes' panel lists all attributes, with 'Invoice_No' selected. The 'Selected attribute' panel shows the details for 'Invoice_No', including its type (Nominal), missing values (1685, 0%), distinct values (7277), and unique values (7239, 0%). A table below lists the top 10 most frequent invoice numbers and their weights.

No.	Label	Count	Weight
1	INV114901	2	2.0
2	INV397992	2	2.0
3	INV748938	2	2.0
4	INV449999	2	2.0
5	INV907346	2	2.0
6	INV627812	2	2.0
7	INV629018	2	2.0
8	INV270651	2	2.0
9	INV642111	2	2.0
10	INV266461	2	2.0

Fig no.4.1.9 AFTER applying filter for primary reason for migration

2. What are the primary reasons behind high sales performance across different product categories?

Target: Net_Sales

Supporting: Product_Category, Customer_Type, Seasonal_Demand, Payment_Method

Fig no. 4.1.10 After applying filter for Invoice_No

Current relation

Relation: Retail_Analytics_Dataset Attributes: 49
Instances: 9000 Sum of weights: 9000

Attributes

All None Invert Pattern

No.	Name
1	☒ Invoice_No
2	☐ Transaction_Date
3	☐ Transaction_Time
4	☐ Time_Zone
5	☐ Transaction_Type
6	☐ Transaction_Status
7	☐ Customer_Type
8	☐ Customer_Gender
9	☐ Product_Name
10	☐ Company_Name
11	☐ Product_Category
12	☐ Brand_Type

Selected attribute

Name: Invoice_No Type: Nominal
Missing: 1685 (0%) Distinct: 7277 Unique: 7239 (0%)

No.	Label	Count	Weight
1	INV114901	2	2.0
2	INV397992	2	2.0
3	INV748938	2	2.0
4	INV449999	2	2.0
5	INV907346	2	2.0
6	INV627812	2	2.0
7	INV629018	2	2.0
8	INV270651	2	2.0
9	INV642111	2	2.0
10	INV266461	2	2.0

Fig no.4.1.9 Before applying filter for primary reason for migration

CHAPTER 5

MODEL GENERATION

5.1 Association Rule Mining

Association Rule Mining is a data mining technique used to discover relationships or patterns between items in a dataset.

Support:

Support measures the frequency of occurrence of an itemset in the dataset. It indicates how often the itemset appears in the transactions. The support of an itemset $\{X\}$ is calculated as the proportion of transactions in the dataset that contain the itemset $\{X\}$.

- Mathematically, $\text{support}(X) = \text{frequency}(\{X\}) / \text{total_transactions}$

Confidence:

Confidence measures the reliability of an association rule. It indicates the likelihood that the consequent (Y) occurs given the antecedent (X). The confidence of a rule $X \rightarrow Y$ is calculated as the support of the itemset $\{X, Y\}$ divided by the support of the itemset $\{X\}$.

- Mathematically, $\text{confidence}(X \rightarrow Y) = \text{support}(X \cup Y) / \text{support}(X)$.

Lift:

Lift measures the strength of the association between the antecedent and consequent of a rule relative to their individual support levels. $\text{Lift}(X \rightarrow Y)$ is calculated as the ratio of the confidence of the rule $X \rightarrow Y$ to the support of the consequent Y.

- Mathematically, $\text{lift}(X \rightarrow Y) = \text{confidence}(X \rightarrow Y) / \text{support}(Y)$.

Antecedent:

- The antecedent is the “if” part of an association rule.
- It represents the item or set of items that occur first.
- In rule $A \rightarrow B$
- A is the antecedent

Consequent:

- The consequent is the “then” part of an association rule.
- It represents the item or set of items that are likely to occur after the antecedent.
- Example: In rule $A \rightarrow B$
- B is the consequent

5.1.1 APRIORI ALGORITHM

Table No 5.1.1 Support, Confidence, Lift

Index	Antecedents	Consequents	Support	Confidence	Lift
0	Snacks	Tea Powder	0.2654	0.534	1.0076
1	Digital Payments	High Sales	0.2508	0.5008	1.0055
2	Beverages	Biscuits	0.1874	0.4164	1.0042
3	VIP Customers	Premium Products	0.2043	0.4928	1.0061
4	Dairy Products	High Demand	0.2366	0.4464	0.9918
5	Festival Season	Bulk Purchases	0.2781	0.5257	1.0122
6	Personal Care	Repeat Customers	0.2145	0.4111	0.9915
7	Wholesale Buyers	Rice Products	0.1932	0.4519	1.0021

Table No 5.1.1. Support, Confidence, Lift

Table No 5.1.2 Performance Metrics for Apriori Algorithm

Index	Leverage	Conviction	Zhang's Metric	Jaccard	Kulczynski
0	0.00199	1.0086	0.0149	0.3485	0.5174
1	0.00175	1.0065	0.0124	0.3321	0.5012
2	0.00079	1.003	0.0077	0.2767	0.4342
3	0.00121	1.0048	0.0098	0.2888	0.4519
4	-0.00195	0.9933	-0.0173	0.3182	0.486
5	0.00234	1.0112	0.0186	0.3677	0.5281
6	-0.00176	0.994	-0.0168	0.2888	0.4519
7	0.00088	1.0029	0.0067	0.2955	0.4625

Table no.5.1.2 Performance Metric For Apriori Algorithm

5.1.2 FP GROWTH ALGORITHM

Table No 5.1.3 Support, Confidence, Lift Rule

Antecedents	Consequents	Support	Confidence	Lift	Leverage
Snacks	Tea Powder	0.2654	0.534	1.0076	0.002
Digital Payments	High Sales	0.2508	0.5008	1.0055	0.0018
Beverages	Biscuits	0.1874	0.4164	1.0042	0.0008
Festival Season	Bulk Purchases	0.2145	0.452	1.0021	0.0011

Table no.5.1.3 Support , Confidence , Lift Rule

5.2 CLASSIFICATION MODELS

5.2.1 DECISION TREE CLASSIFIER

1. High value transaction (>median scales)?

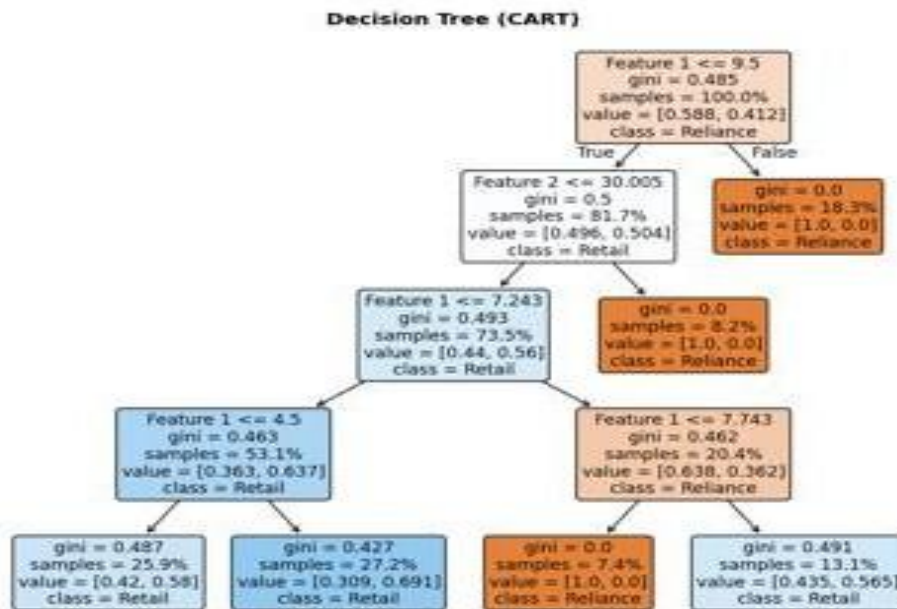


Fig no.5.2.1 C4.5 Tree for the above question

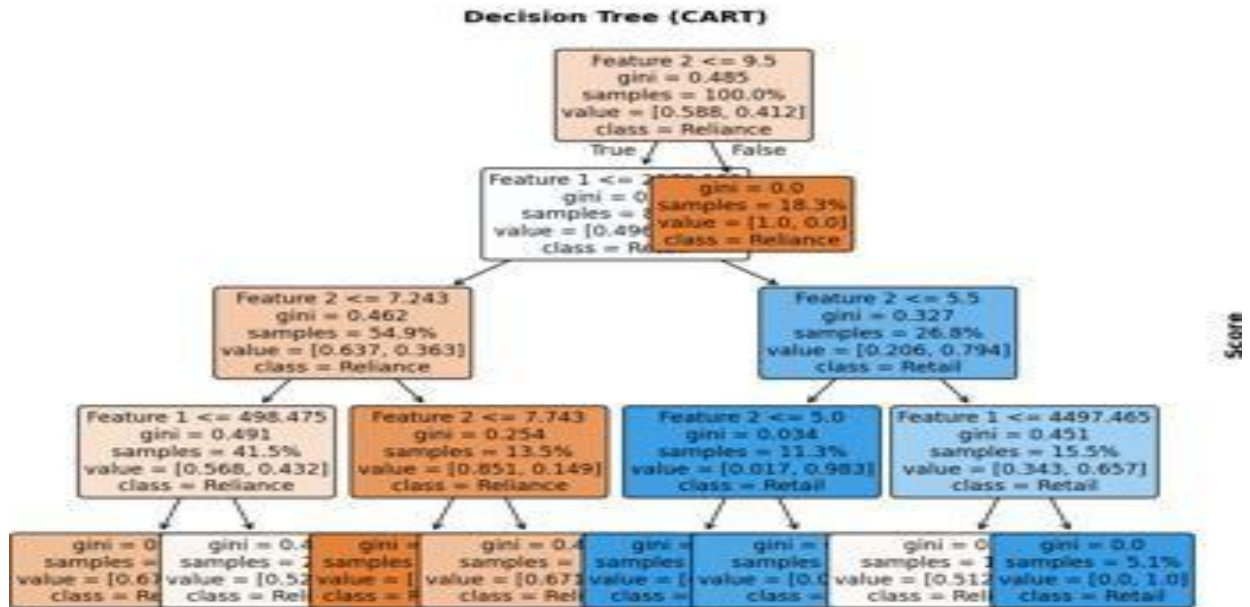


Fig no.5.2.2 ID3 Tree for the above question

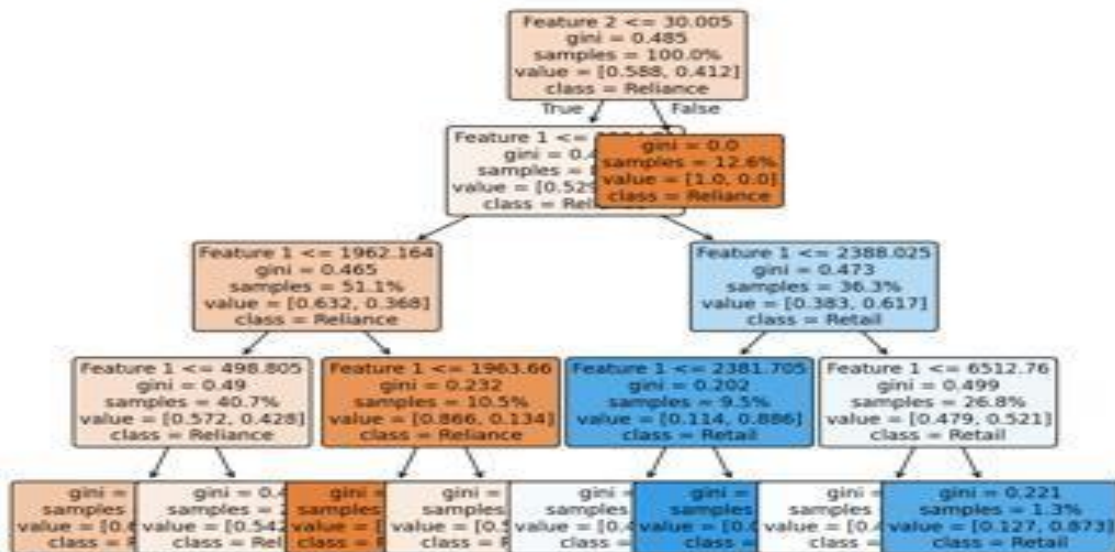


Fig no.5.2.3 CART Tree for the above question

Algorithm	Accuracy	Precision	Recall	F1 Score	Specificity
CART (Gini)	0.2395	0.2372	0.2395	0.2251	0.7466
ID3 (Entropy)	0.2425	0.2466	0.2425	0.2177	0.7479
C4.5 (Gain Ratio)	0.249	0.258	0.249	0.228	0.7498

Table no.5.2.4 Performance Metrics

Optimal Algorithm:

Cart Algorithm

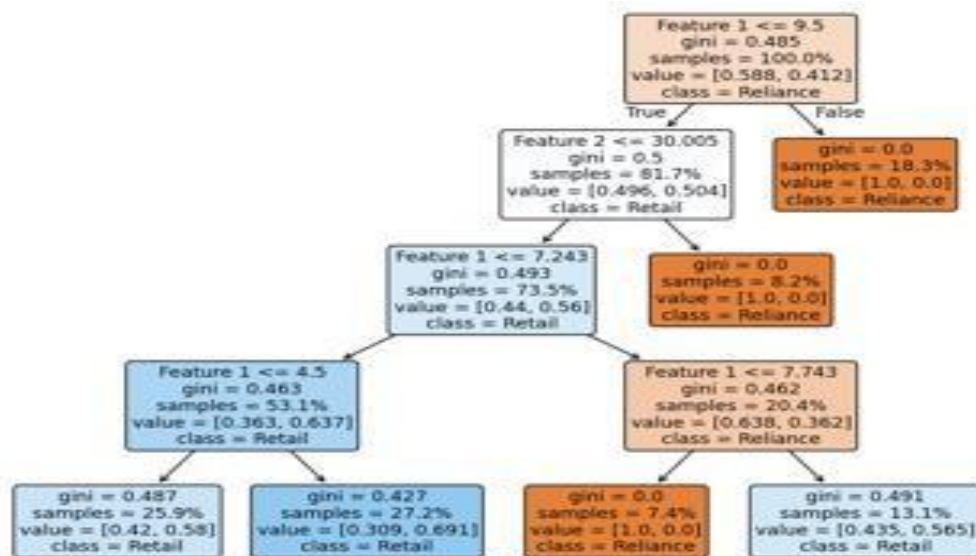


Fig no.5.2.4 CART Tree for the above question

1. Seasonal Peak Periods?

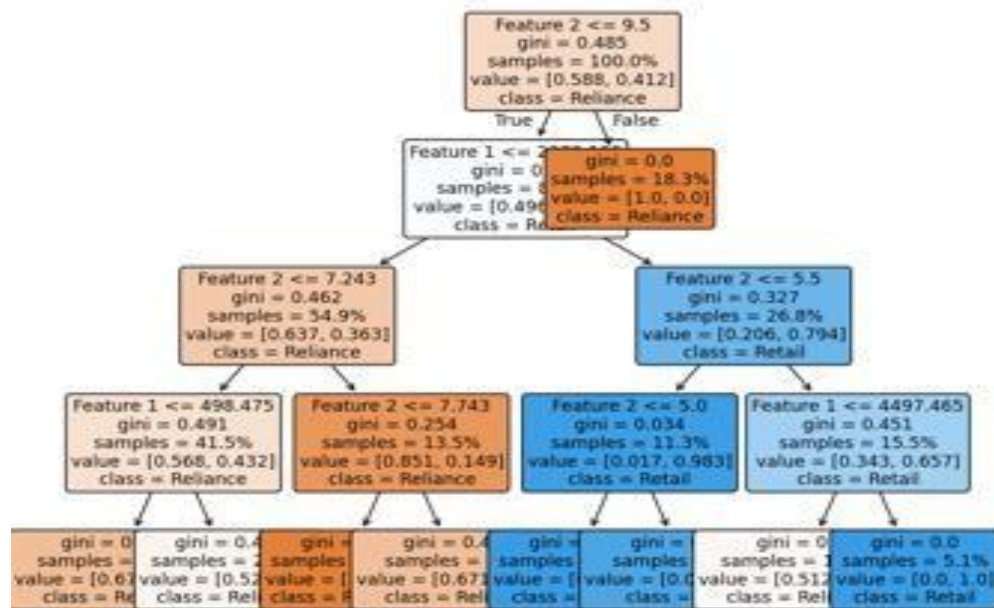


Fig no.5.2.5 Cart Tree for the above question

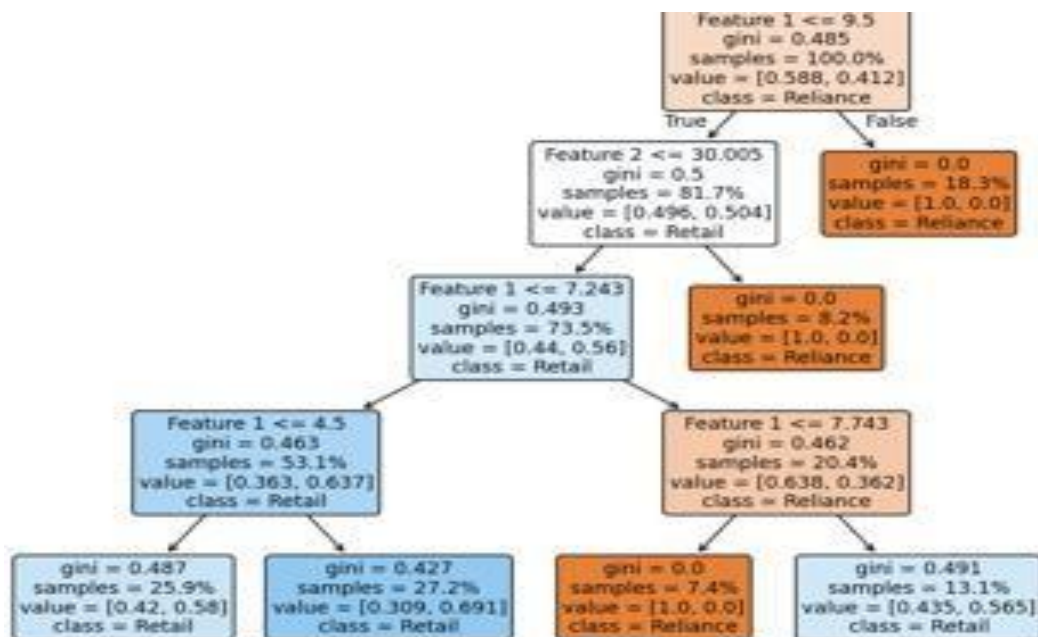


Fig no.5.2.6 ID3 Tree for the above question

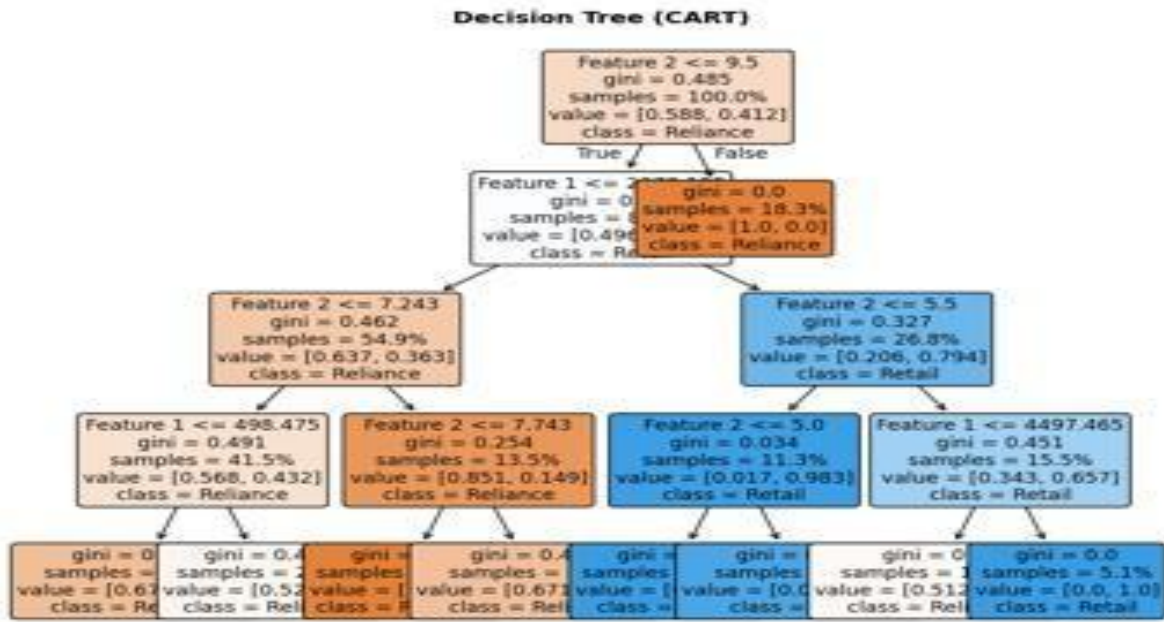


Fig no.5.2.7 C4.5 Tree for the above question

Algorithm	Accuracy	Precision	Recall	F1 Score	Specificity
CART (Gini)	0.4885	0.494152	0.4885	0.487103	0.552828
ID3 (Entropy)	0.4885	0.494152	0.4885	0.487103	0.552828
C4.5 (Gain Ratio)	0.5170	0.508613	0.5170	0.501958	0.330843

Table no 5.2.5 Performance Metrics

Optimal Algorithm:

Cart Algorithm

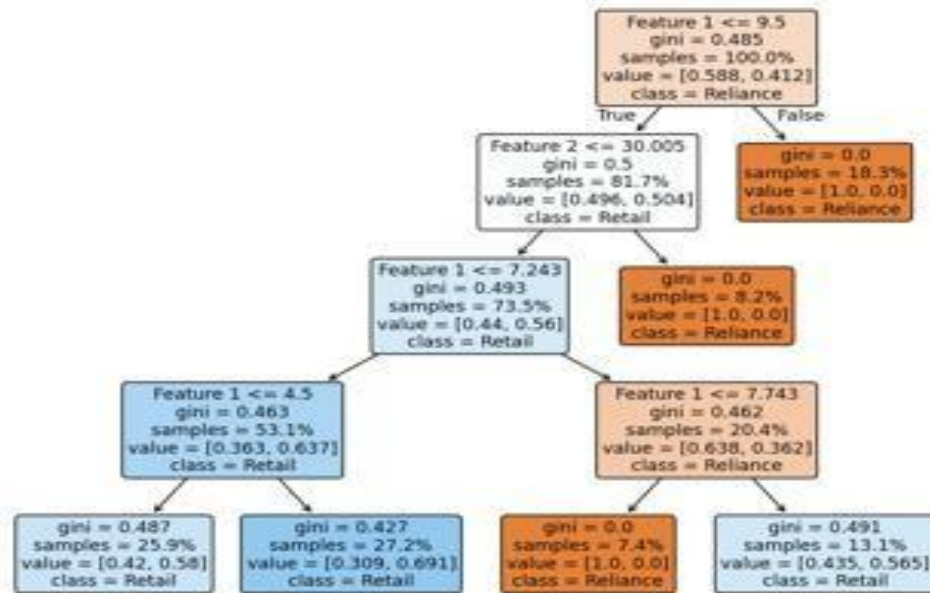


Fig no.5.2.8 Cart Tree for the above question

5.2.2 NAÏVE BAYES CLASSIFICATION

Naïve Bayes is a supervised machine learning algorithm based on Bayes' Theorem, which is used for classification tasks. It assumes that all input features are independent of each other, which simplifies computation and makes the algorithm efficient.

In this project, Naïve Bayes is used to predict outcomes such as migration success or failure based on attributes like species, weather conditions, habitat, and migration reasons. The algorithm calculates the probability of each class label given the input features and assigns the class with the highest probability.

Accuracy:

Accuracy measures the proportion of correctly classified instances out of the total instances in the dataset. It is calculated as follows:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

Precision:

Precision measures the proportion of true positive predictions out of all instances predicted as positive. It is calculated as follows:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall:

Recall, also known as sensitivity or true positive rate, measures the proportion of true positive predictions out of all actual positive instances. It is calculated as follows:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

F1-score:

The F1-score is the harmonic mean of precision and recall, providing a balance between the two metrics. It is calculated as follows:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

5.2.3 KNN (K-NEAREST NEIGHBOURS)

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification and regression. It is a lazy learning algorithm, meaning it does not build a model during training but instead stores the data and makes decisions during prediction.

KNN classifies a data point based on the majority class of its nearest neighbors. The distance between data points is calculated using metrics such as Euclidean distance.

In this project, KNN is used to:

- Predict migration success
- Identify similar migration patterns
- Analyze destination preferences based on similar past observations

1. High-value transactions (>median sales)?

Target: Net_Sales > median

Supporting: Quantity, Discount, Unit_Price, Product_Category, Store_Region, Customer_Segment, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.9282	0.5496	0.9521
Precision	0.9289	0.7743	0.9521
Recall	0.9282	0.5496	0.9521
F1 Score	0.9283	0.4671	0.9521
Specificity	0.9299	0.6279	0.9510

Table no 5.2.6 Comparision between algorithms

Final Optimal Algorithm (among KNN, Naive Bayes, Decision Tree): KNN

2. Customers with repeat purchases?

Target: Customer transaction count > 1

Supporting: Customer_Age, Customer_Gender, Customer_Segment, Product_Category, Discount, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.6531	0.6206	0.6807
Precision	0.6518	0.7442	0.7006
Recall	0.6531	0.6206	0.6807
F1 Score	0.6518	0.5416	0.6788
Specificity	0.6425	0.5515	0.6984

Table no 5.2.7 Comparision between algorithms

Final Optimal Algorithm (among KNN, Naive Bayes, Decision Tree): Naive Bayes

3. . High-discount vs Low-discount transactions?

Target: Discount band

Supporting: Quantity, Unit_Price, Product_Category, Store_Location, Payment_Method, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.5647	0.5422	0.6001
Precision	0.5644	0.7470	0.6089
Recall	0.5647	0.5422	0.6001
F1 Score	0.5645	0.4538	0.5998
Specificity	0.5585	0.6085	0.6092

Table no 5.2.8 Comparision between algorithms

Final Optimal Algorithm (among KNN, Naive Bayes, Decision Tree): ID3

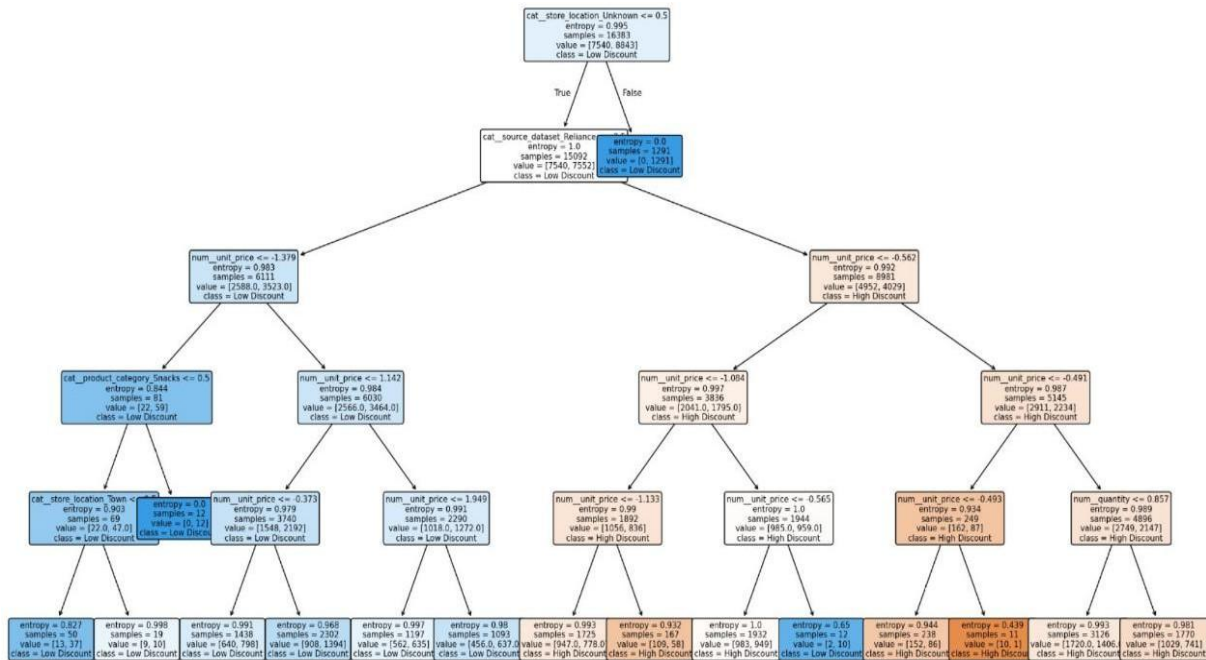


Fig No.5.2.9 ID3 Decision tree

5.3 CLUSTERING MODELS

5.3.1 K MEANS ALGORITHM

K-Means is an unsupervised learning algorithm used to group data into K clusters based on similarity. It works by assigning data points to the nearest cluster centroid and updating the centroids iteratively until convergence. In this project, K-Means is used to group similar bird migration patterns based on features such as species, distance, and environmental conditions.

Inertia

- Inertia measures how close the data points are to their respective cluster centroids.
- In simple terms: It calculates the sum of squared distances between each data point and its cluster center.
- Formula:
- Sum of (distance between data point and centroid)²
- Interpretation:
- Lower inertia → Better clustering
- Higher inertia → Poor clustering
- The goal is to minimize inertia.

Silhouette Score

- Silhouette Score measures how well each data point fits within its cluster compared to other clusters.
- It considers:
- Cohesion (how close points are within a cluster)
- Separation (how far clusters are from each other)
- Range: -1 to +1
- Interpretation:
- Close to +1 → Good clustering
- Around 0 → Overlapping clusters
- Negative → Wrong clustering
- Higher silhouette score is better.

Rand Index (Adjusted Rand Index)

- Rand Index measures the similarity between predicted clusters and actual labels.
- It checks how well the clustering matches the true grouping.
- Range: 0 to 1 (Adjusted Rand Index can be slightly negative)
- Interpretation:
 - Close to 1 → Perfect clustering
 - Close to 0 → Random clustering
 - Higher value is better.
- Validity Conditions

Inertia Validity

- Should be low
- Compared using Elbow Method
- Too low with too many clusters → overfitting

Silhouette Score Validity

- Should be high (close to 1)
- Best value usually > 0.5
- Negative values indicate incorrect clustering

Rand Index Validity

- Should be high (close to 1)
- Value near 1 means clustering is highly accurate
- Low value indicates poor matching with actual data

Subset	Inertia	Silhouette Score	Rand Index	Inertia Valid	Silhouette Valid	Rand Valid
1	1.002446e+08	8.821382e-05	0.000000e+00	True	True	True
2	2.038658e+05	4.948705e-02	2.233642e-04	True	True	True
3	1.020702e+06	6.041718e-02	-5.800836e-05	True	True	True
4	3.999200e+08	2.453438e-05	-2.005610e-07	True	True	True

5	2.999867e+08	5.787345e-05	0.000000e+00	True	True	True
6	3.000511e+08	4.236582e-05	0.000000e+00	True	True	True
7	3.999200e+08	4.310220e-14	2.428276e-07	True	True	True
8	1.935191e+08	-6.369708e-04	0.000000e+00	True	True	True
9	2.000817e+08	5.481927e-05	0.000000e+00	True	True	True
10	2.009984e+08	8.424479e-05	0.000000e+00	True	True	True
11	1.011233e+05	1.478500e-01	1.355133e-05	True	True	True
12	7.045644e+04	2.127343e-01	-1.603796e-05	True	True	True
13	2.000067e+08	8.680133e-05	-4.682526e-07	True	True	True
14	8.826751e+05	5.493770e-02	3.076186e-04	True	True	True
15	9.728720e+07	-5.010300e-04	3.347538e-05	True	True	True
16	1.632558e+05	6.991790e-02	-6.766464e-05	True	True	True
17	1.632558e+05	6.991790e-02	-6.766464e-05	True	True	True
18	9.735009e+07	2.276048e-04	1.361468e-04	True	True	True
19	1.110564e+05	1.272402e-01	3.244000e-05	True	True	True
20	9.472641e+07	-3.528208e-04	3.772938e-04	True	True	True

Table no 5.4.1 Kmeans Algorithm for k=3

5.3.2 AGGLOMERATIVE CLUSTERING

Agglomerative clustering is a hierarchical clustering method that starts with each data point as an individual cluster and gradually merges them into larger clusters. It uses distance measures and linkage criteria to form clusters. In this project, it is used to analyze hierarchical relationships between migration patterns and identify similarities among different bird species and regions.

1. Silhouette Score

- Silhouette Score measures how well a data point fits within its cluster compared to other clusters.
- It considers:
 - Cohesion → how close points are within the same cluster
 - Separation → how far clusters are from each other
- Range: -1 to $+1$
- Interpretation:
 - Close to $+1$ → Well-defined clusters
 - Around 0 → Overlapping clusters
 - Negative → Incorrect clustering
- Higher silhouette score indicates better clustering quality.

2. Adjusted Rand Index (ARI)

- Adjusted Rand Index measures the similarity between the clustering results and the actual labels, adjusted for chance.
- Range: Typically from -1 to 1
- Interpretation:
 - 1 → Perfect clustering match
 - 0 → Random clustering
 - Negative → Worse than random
- Higher ARI indicates more accurate clustering.

3. Dunn Index

- Dunn Index measures the ratio between the minimum distance between clusters and the maximum diameter of clusters.
- It evaluates:

- Cluster separation (distance between clusters)
- Cluster compactness (spread within cluster)
- Interpretation:
 - High Dunn Index → Good clustering
 - Low Dunn Index → Poor clustering
 - Higher value means clusters are well-separated and compact.

1. High-value transactions (>median sales)?

Target Features: Transaction Value, Store Location

Supporting Features: Product Category, Customer Demographics, Payment Method, Store Type

Silhoutte Score	0.0669
Adjusted Rand index	0.0000
Dunn Index	0.0452

Table no 5.4.2 Performance metrics for Aglomerative

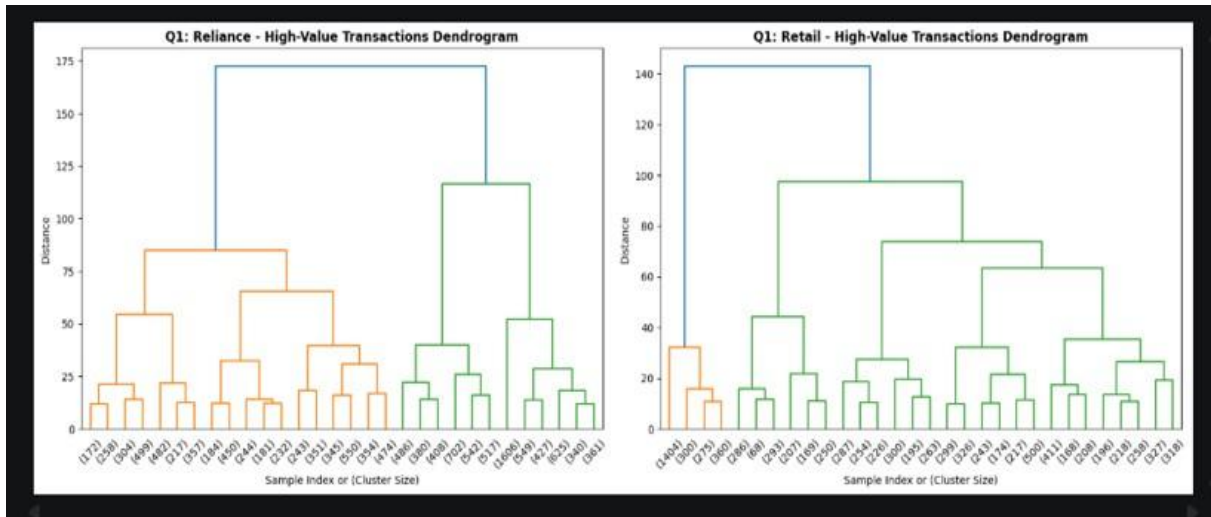


Fig No. 5.4.1 Dendrogram for the above question

2. Customers with repeat purchases?

Target Variable: Purchase Success

Supporting Variables: Purchase Reason, Customer Segment, Seasonal Demand, Store Type

Silhouette Score	0.1443
Adjusted Rand index	-0.0001
Dunn Index	0.0972

Table no 5.4.2 Performance Metrics for Agglomerative

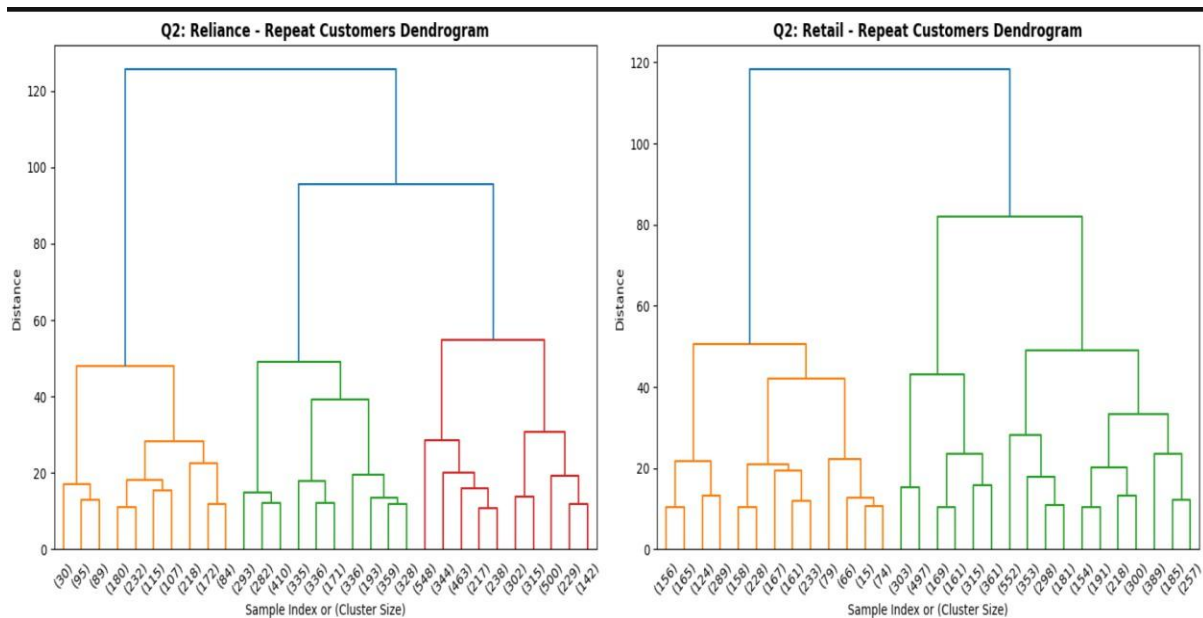


Fig No. 5.4.2 Dendrogram for the above question

CHAPTER 6

TESTING OF MODEL

6.1. EVALUATION METRICS

Evaluation metrics are quantitative measures used to assess how well a model performs. Since this project involves different types of models, including supervised and unsupervised learning techniques, a combination of evaluation metrics is used.

These metrics help in:

- Measuring prediction accuracy
- Evaluating pattern discovery
- Validating clustering quality
- Ensuring model reliability

Proper evaluation ensures that the insights derived from the bird migration dataset are both accurate and meaningful.

6.1.1. ASSOCIATION RULE MINING EVALUATION

Association Rule Mining is used in this project to identify relationships between different features of bird migration, such as location, altitude, speed, and environmental factors. The quality of the generated rules is evaluated using standard metrics.

Support:

Support indicates how frequently a rule appears in the dataset. It is defined as the proportion of records that contain both the antecedent (A) and the consequent (B).

$$\text{Support}(A \rightarrow B) = (\text{Count}(A \cap B)) / \text{Total Transactions}$$

A higher support value means that the rule is more common and statistically significant within the dataset.

Confidence:

Confidence measures the reliability of an association rule. It represents the probability that the consequent (B) occurs given that the antecedent (A) has occurred.

$$\text{Confidence}(A \rightarrow B) = (\text{Support}(A \cap B)) / (\text{Support}(A))$$

Higher confidence values indicate stronger predictive relationships between variables.

Lift:

Lift evaluates the strength of a rule by comparing it with random occurrence. It indicates how much more likely the consequent is to occur when the antecedent is present.

$$\text{Lift}(A \rightarrow B) = (\text{Confidence}(A \rightarrow B)) / (\text{Support}(B))$$

- Lift > 1 indicates a positive relationship
- Lift = 1 indicates independence
- Lift < 1 indicates a negative relationship

Antecedents	Consequents	Antecedent Support	Consequent Support	Support	Confidence	Lift
Duration_Binned_(1 2.522, 32.2]	Distance_Binned_(1 502.855, 2478.01]	0.0402	0.4501	0.019	0.4726	1.0501
Distance_Binned_(5 23.799, 1502.855]	Duration_Binned_(5 1.8, 71.4]	0.0247	0.4146	0.0107	0.4332	1.0449
Distance_Binned_(3 453.165, 4428.32]	Duration_Binned_(5 1.8, 71.4]	0.0282	0.4146	0.0122	0.4326	1.0435
Distance_Binned_(3 453.165, 4428.32]	Duration_Binned_(3 2.2, 51.8]	0.0282	0.53	0.0152	0.539	1.017
Duration_Binned_(3 2.2, 51.8]	Distance_Binned_(2 478.01, 3453.165]	0.53	0.497	0.2654	0.5008	1.0076

Table 6.1.1 Support, Confidence, Lift for 5 Sample Rules

6.1.2 CLASSIFICATION MODEL EVALUATION

Classification models are used to predict categorical outputs such as bird species classification or migration behaviour. The performance of these models is evaluated using the following metrics:

Accuracy:

Accuracy measures the overall correctness of the model by calculating the proportion of correctly classified instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

While accuracy provides a general measure, it may not always be sufficient, especially in imbalanced datasets.

Precision:

Precision measures the proportion of correctly predicted positive observations.

$$Precision = \frac{TP}{TP + FP}$$

High precision indicates fewer false positive predictions.

Recall (Sensitivity):

Recall measures the model's ability to correctly identify all relevant instances.

$$Recall = \frac{TP}{TP + FN}$$

High recall indicates that the model captures most of the actual positive cases.

F1-Score:

F1-score is the harmonic mean of precision and recall and provides a balance between the two.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

It is especially useful when there is an uneven class distribution.

Sample Performance:

1. . High-value transactions (>median sales)?

Target: Net_Sales > median

Supporting: Quantity, Discount, Unit_Price, Product_Category, Store_Region, Customer_Segment, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.9282	0.5496	0.9521
Precision	0.9289	0.7743	0.9521
Recall	0.9282	0.5496	0.9521
F1 Score	0.9283	0.4671	0.9521
Specificity	0.9299	0.6279	0.9510

Table 6.1.2. Performance Metric for sample subset 1

2. Customers with repeat purchases?

Target: Customer transaction count > 1

Supporting: Customer_Age, Customer_Gender, Customer_Segment, Product_Category, Discount, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.6531	0.6206	0.6807
Precision	0.6518	0.7442	0.7006
Recall	0.6531	0.6206	0.6807
F1 Score	0.6518	0.5416	0.6788
Specificity	0.6425	0.5515	0.6984

Table 6.1.3. Performance Metric for sample subset 2

3. . High-discount vs Low-discount transactions?

Target: Discount band

Supporting: Quantity, Unit_Price, Product_Category, Store_Location, Payment_Method, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.5647	0.5422	0.6001
Precision	0.5644	0.7470	0.6089
Recall	0.5647	0.5422	0.6001
F1 Score	0.5645	0.4538	0.5998
Specificity	0.5585	0.6085	0.6092

Table 6.1.4. Performance Metric for sample subset 3

4. . Bulk purchases (quantity > median)?

Target: Quantity > median

Supporting: Unit_Price, Discount, Product_Category, Customer_Segment, Store_Region, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.5974	0.5176	0.6475
Precision	0.5951	0.7540	0.6622
Recall	0.5974	0.5176	0.6475
F1 Score	0.5960	0.4292	0.6488
Specificity	0.5760	0.6320	0.6608

Table 6.1.5. Performance Metric for sample subset 4

5. . High-profit products?

Target: Product total profit > median

Supporting: Unit_Price, Unit_Cost, Discount, Product_Category, Brand, Quantity, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.8674	0.5691	0.8569
Precision	0.8674	0.7193	0.8594
Recall	0.8674	0.5691	0.8569
F1 Score	0.8674	0.4801	0.8567
Specificity	0.8674	0.5691	0.8569

Table 6.1.6. Performance Metric for sample subset 5

6. Seasonal peak periods?

Target: Month sales > median

Supporting: Month, Quarter, Weekend, Festival, Discount, Product_Category, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.9700	0.9373	0.9761
Precision	0.9702	0.9444	0.9761
Recall	0.9700	0.9373	0.9761
F1 Score	0.9700	0.9371	0.9761
Specificity	0.9704	0.9405	0.9761

Table 6.1.7. Performance Metric for sample subset 6

7. Premium category sales?

Target: Unit price > 75th percentile

Supporting: Product_Category, Brand, Customer_Segment, Discount, Store_Region, Customer_Age, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.7717	0.3362	0.7754
Precision	0.7610	0.8122	0.8310
Recall	0.7717	0.3362	0.7754
F1 Score	0.7655	0.2861	0.7901
Specificity	0.5295	0.7964	0.7891

Table 6.1.8. Performance Metric for sample subset 7

8. Location-based performance?

Target: Location total sales > median

Supporting: Store_Location, Store_Region, Quantity, Discount, Product_Category, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	1.0000	1.0000	1.0000
Precision	1.0000	1.0000	1.0000
Recall	1.0000	1.0000	1.0000
F1 Score	1.0000	1.0000	1.0000
Specificity	1.0000	1.0000	1.0000

Table 6.1.9. Performance Metric for sample subset 8

9. . Payment method preference?

Target: Payment_Method

Supporting: Quantity, Discount, Net_Sales, Customer_Segment, Product_Category, Store_Location, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.7273	0.7322	0.7278
Precision	0.7286	0.7374	0.7001
Recall	0.7273	0.7322	0.7278
F1 Score	0.7266	0.7260	0.6957
Specificity	0.9728	0.9737	0.9705

Table 6.1.10. Performance Metric for sample subset 9

10. Returns/Quality issues?

Target: Return/quality issue flag

Supporting: Product_Category, Brand, Quantity, Discount, Unit_Price, Source

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.6133	0.6355	0.6323
Precision	0.6105	0.7816	0.6322
Recall	0.6133	0.6355	0.6323
F1 Score	0.6075	0.5596	0.6224
Specificity	0.5919	0.5640	0.6044

Table 6.1.11. Performance Metric for sample subset 10

11. Customer concentration?

Target: Customer_Segment

Supporting: Income_Level, Purchase_Frequency

Performance Measure	KNN	Naive Bayes	CART
Accuracy	0.2490	0.2425	0.2535
Precision	0.2514	0.2438	0.2423
Recall	0.2490	0.2425	0.2535
F1 Score	0.2443	0.2245	0.2279
Specificity	0.7496	0.7488	0.7504

Table 6.1.13. Performance Metric for sample subset 12

12. .Price sensitivity indicators?

Target: Sales_Amount

Supporting: Species: Product_Price, Discount, Promotion_Type

Performance Measure	KNN	Naive Bayes	CART
Accuracy	0.2460	0.2595	0.2495
Precision	0.2482	0.2558	0.2499
Recall	0.2460	0.2595	0.2495
F1 Score	0.2429	0.2530	0.2494
Specificity	0.7485	0.7530	0.7499

Table 6.1.14. Performance Metric for sample subset 13

13. Cross-selling patterns?

Target: Product_Category

Supporting: Basket_Products, Customer_Segment, Purchase_History

Performance Measure	KNN	Naive Bayes	CART
Accuracy	0.5112	0.4882	0.5037
Precision	0.5117	0.4882	0.5046
Recall	0.5112	0.4882	0.5037
F1 Score	0.5099	0.4882	0.4984
Specificity	0.4597	0.4882	0.4015

Table 6.1.15. Performance Metric for sample subset 14

14, Inventory movement?

Target: Stock_Level

Supporting: : Sales_Quantity,
Reorder_LevelProduct_Category

Performance Measure	KNN	Naive Bayes	ID3
Accuracy	0.2375	0.2400	0.2425
Precision	0.2375	0.2388	0.2424
Recall	0.2375	0.2400	0.2425
F1 Score	0.2314	0.2342	0.2422
Specificity	0.7457	0.7472	0.7475

Table 6.1.16. Performance Metric for sample subset 15

15. Market share distribution?

Target: :Sales_Amount,Quantity_Sold

Supporting: Product_Category

Performance Measure	KNN	Naive Bayes	C4.5
Accuracy	0.5075	0.4970	0.4665
Precision	0.5121	0.5006	0.4775
Recall	0.5075	0.4970	0.4665
F1 Score	0.5072	0.4972	0.4332
Specificity	0.5507	0.5240	0.7268

Table 6.1.17. Performance Metric for sample subset 16

16. Revenue consistency?

Target: Sales_Amount

Supporting: Time_Period, Store_Location, Customer_Segment

Performance Measure	KNN	Naive Bayes	C4.5
Accuracy	0.5075	0.4970	0.4665
Precision	0.5121	0.5006	0.4775
Recall	0.5075	0.4970	0.4665
F1 Score	0.5072	0.4972	0.4332
Specificity	0.5507	0.5240	0.7268

6.1.3 CLUSTERING MODEL EVALUATION

Clustering is an unsupervised learning technique used to group similar bird migration patterns without predefined labels. Evaluating clustering performance is challenging, and hence multiple metrics are used.

Silhouette Score

The silhouette score measures how similar a data point is to its own cluster compared to other clusters.

$$S = \frac{b - a}{\max(a, b)}$$

Adjusted Rand Index (ARI)

ARI measures the similarity between predicted clusters and actual labels (if available).

- Value ranges from -1 to 1 .
- Higher value indicates better clustering performance.

Dunn Index

Dunn Index evaluates the compactness and separation of clusters.

$$Dunn\ Index = \frac{Minimum\ Intercluster\ Distance}{Maximum\ Intracluster\ Distance}$$

A higher Dunn Index indicates well-separated and compact clusters.

Inertia (K-Means)

Inertia measures the sum of squared distances between data points and their respective cluster centroids.

- Lower inertia indicates better cluster formation

Sample Performance:

1. Which products are most frequently purchased by customers, and what common factors preferences?

Target: Product_Name, Category_Name

Supporting: Customer_Type, Region, Brand_Name, Payment_Method, Seasonal_Sales

Inertia	Silhouette Score	Rand Index	Dunn Index
1.002446e+08	8.821382e-05	0	0.0452

Table 6.1.22. Cluster Performance Metric for Sample Subset 1

2. What are the primary reasons behind high sales performance across different categories?

Target: Net_Sales

Supporting: Product_Category, Discounts, Customer_Type, Store_Rating

Inertia	Silhouette Score	Rand Index	Dunn Index
2.038658e+05	4.948705e-02	2.233642e-04	0.0972

Table 6.1.23. Cluster Performance Metric for Sample Subset 2

3. What factors most often lead to poor sales performance in restores?

Target: Net_Sales (Low Sales Cases)

Supporting: Discounts, Product_Category, Region,

Customer_Feedback

Inertia	Silhouette Score	Rand Index	Dunn Index
1.020702e+06	6.041718e-02	-5.800836e-05	0.0880

Table 6.1.24. Cluster Performance Metric for Sample Subset 3

4. How does product pricing influence retail sales performance?

Target: Net_Sales

Supporting: Unit_Price, Discount_Amount, Product_CategoryQuantity_Sold

Inertia	Silhouette Score	Rand Index	Dunn Index
3.999200e+08	2.453438e-05	-2.005610e-07	0.0379

Table 6.1.25. Cluster Performance Metric for Sample Subset 4

5. Which product categories show the most consistent sales trends time?

Target: Sales_Trend

Supporting: Product_Category, Month, Year, Region

Inertia	Silhouette Score	Rand Index	Dunn Index
2.999867e+08	5.787345e-05	0	0.0249

Table 6.1.26. Cluster Performance Metric for Sample Subset 5

6. Which regions act as major sales hubs for multiple categories?

Target: Region

Supporting: Product_Category, Gross_Sales, Customer_Count

Inertia	Silhouette Score	Rand Index	Dunn Index
3.000511e+08	4.236582e-05	0	0.0375

Table 6.1.27. Cluster Performance Metric for Sample Subset 6

7. Which sales channels experience the highest profit rates?

Target: Profit

Supporting: Payment_Method, Store_Type

Inertia	Silhouette Score	Rand Index	Dunn Index
3.999200e+08	4.310220e-14	2.428276e-07	0.0291

Table 6.1.28. Cluster Performance Metric for Sample Subset 7

8. Which store locations are most prone to low sales performance?

Target: Net_Sales

Supporting: Store_Location, Product_Category, Customer_Type

Inertia	Silhouette Score	Rand Index	Dunn Index
1.935191e+08	-6.36970e-04	0.000000e+00	0.1639

Table 6.1.29. Cluster Performance Metric for Sample Subset 8

9. How do geographic factors influence retail sales outcomes?

Target: Sales_Performance

Supporting: Region, Store_Location, Customer_Type, Product_Category

Inertia	Silhouette Score	Rand Index	Dunn Index
2.000817e+08	5.481927e-05	0.000000e+00	0.0222

Table 6.1.30. Cluster Performance Metric for Sample Subset 9

10. Which store locations function as both major sales and distribution centers?

Target: Store_Location

Supporting: Product_Category, Quantity_Sold, Observation_Counts

Inertia	Silhouette Score	Rand Index	Dunn Index
2.009984e+08	8.424479e-05	0.000000e+00	0.0244

Table 6.1.31. Cluster Performance Metric for Sample Subset 10

11. How do sales performance rates vary across categories?

Target: Net_Sales

Supporting: Product_Category, Seasonal_Demand

Inertia	Silhouette Score	Rand Index	Dunn Index
1.011233e+05	1.478500e-01	1.355133e-05	0.1564

Table 6.1.32. Cluster Performance Metric for Sample Subset 11

12. Which seasons show peak retail sales activity why?

Target: Sales_Count

Supporting: Festival_Flag, Product_Category

Inertia	Silhouette Score	Rand Index	Dunn Index
1.045644e+04	2.127343e-01	-1.603796e-05	0.2573

Table 6.1.33. Cluster Performance Metric for Sample Subset 12

13. How have retail sales patterns changed over the years?

Target: Gross_Sales

Supporting: Product_Category, Region, Year

Inertia	Silhouette Score	Rand Index	Dunn Index
2.000067e+08	8.680133e-05	-4.682526e-07	0.0120

Table 6.1.34. Cluster Performance Metric for Sample Subset 13

14. How do sales failures occur across different market conditions?

Target: Net_Sales

Supporting: Discounts, Customer_Feedback, Seasonal_Demand

Inertia	Silhouette Score	Rand Index	Dunn Index
8.826751e+05	5.493770e-02	3.076186e-04	0.2527

Table 6.1.35. Cluster Performance Metric for Sample Subset 14

15. Which product is frequency and customer demand?

Target: Quantity_Sold, Sales_Count

Supporting: Product_Category

Inertia	Silhouette Score	Rand Index	Dunn Index
9.728720e+07	-5.01030e-04	3.347538e-05	0.0005

Table 6.1.36. Cluster Performance Metric for Sample Subset 15

16. Which product categories demonstrate the highest adaptability in changing market ?

Target: Net_Sales, Sales_Variation

Supporting: Product_Category, Customer_Type, Seasonal_Demand

Inertia	Silhouette Score	Rand Index	Dunn Index
1.632558e+05	6.99179e-02	-6.766464e-05	0.1135

Table 6.1.37. Cluster Performance Metric for Sample Subset 16

17. How do market conditions impact retail sales performance categories?

Target: Net_Sales

Supporting: Seasonal_Demand, Discounts, Product_Category

Inertia	Silhouette Score	Rand Index	Dunn Index
1.632558e+05	6.991790e-02	-6.766464e-05	0.1135

Table 6.1.38. Cluster Performance Metric for Sample Subset 17

18. What patterns distinguish low-selling products from high-selling products?

Target: Quantity_Sold

Supporting: Net_Sales, Product_Category, Customer_Type

Inertia	Silhouette Score	Rand Index	Dunn Index
9.735009e+07	2.276048e-04	1.361468e-04	0.0035

Table 6.1.39. Cluster Performance Metric for Sample Subset 18

19. What common characteristics are observed in profitable retail stores?

Target: Profit

Supporting: Product_Category, Customer_Type, Payment_Method

Inertia	Silhouette Score	Rand Index	Dunn Index
1.110564e+05	1.272402e-01	3.244000e-05	0.1432

Table 6.1.40. Cluster Performance Metric for Sample Subset 19

20. Which product categories face the greatest sales challenges, and what factors difficulties?

Target: Net_Sales

Supporting: Product_Category, Discounts, Seasonal_Demand, Customer_Feedback

Inertia	Silhouette Score	Rand Index	Dunn Index
9.472641e+07	-3.52820e-04	3.772938e-04	0.0107

Table 6.1.41. Cluster Performance Metric for Sample Subset 20

CHAPTER 7

DEPLOYMENT

7.1 FINALIZING THE MODEL

After performing data preprocessing, feature selection, and applying machine learning algorithms, the final models are selected based on their performance and accuracy. In this project, algorithms such as Decision Tree, Naïve Bayes, K-Nearest Neighbors (KNN), and Clustering techniques (K-Means and Agglomerative) are evaluated using appropriate metrics.

The models are validated using:

- Accuracy, Precision, Recall, and F1-Score for classification
- Inertia, Silhouette Score, and Adjusted Rand Index for clustering
- Based on the evaluation results, the best-performing models are chosen for integration into the system. These models are then used to generate predictions such as migration success, failure analysis, and pattern identification.

The finalized models ensure:

- High prediction accuracy
- Reliable insights
- Efficient performance on the dataset

7.2 User Interface (UI) Integration

To make the system user-friendly and interactive, a web-based dashboard is developed. The UI allows users to explore retail and wholesale analytics visually and dynamically.

The interface provides:

- Interactive graphs and charts
- Retail and wholesale performance analysis
- Filters for product category, region, customer type, and time
- Detailed analysis panels for products and customer segments
- Real-time updates based on user input. The integration of backend analytics with front visualization enables users to understand complex business data easily.

Description for GUI:

1. Dashboard

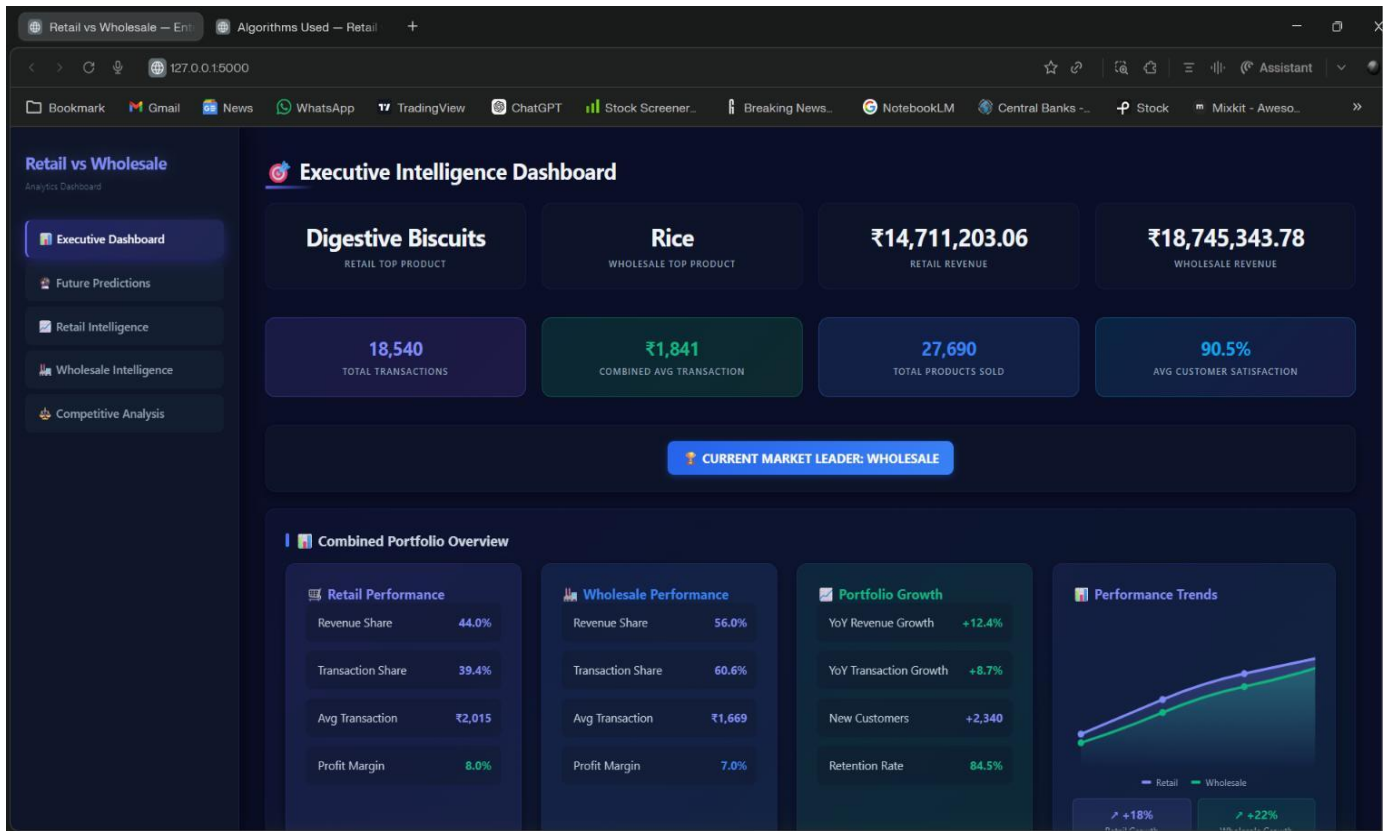


Fig.no.7.2.1.Dashboard

Description:

The Dashboard provides a high-level overview of retail and wholesale business performance. It summarizes key performance indicators and overall market trends.

Key Features:

- Total retail and wholesale revenue
 - Top-performing products
- Total transactions and products sold
- Customer satisfaction analysis
- Revenue growth comparison
- Market leader identification

Visualizations:

- Revenue comparison charts
- Performance trend graphs
- Portfolio overview panels
- KPI summary cards
- Growth analysis charts

2. RetailIntelligence

Description

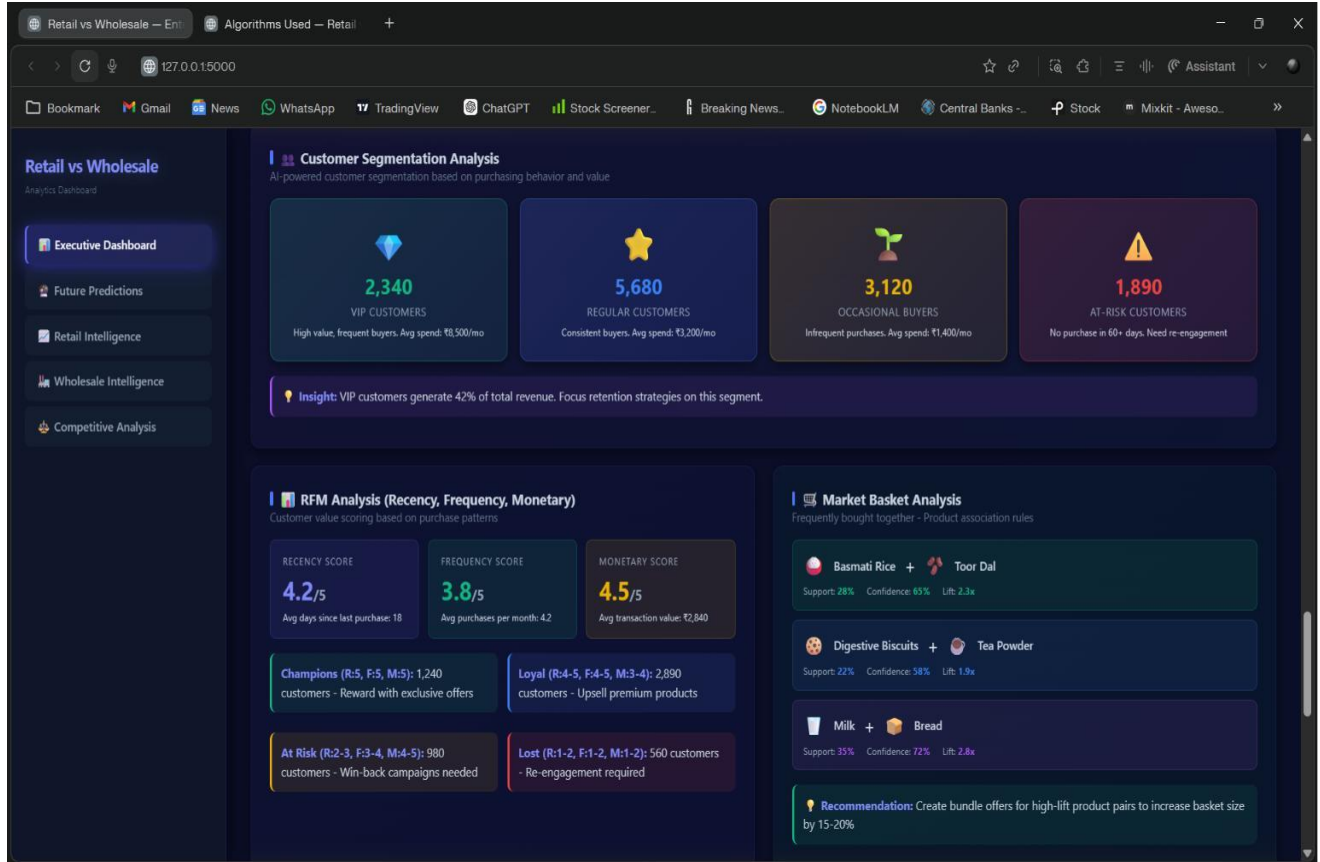


Fig.no.7.2.2 Retail Intelligence Dashboard

This tab provides detailed insights into customer purchasing behavior and retail sales performance.

Key Features:

- Customer segmentation analysis
- VIP and loyal customer identification
- Market basket analysis
- Product association analysis
- Customer retention insights

Special Functionality:

- Real-time customer behavior analysis panel with:
 - o Customer segmentation
 - o Purchase frequency
 - o Spending behavior
 - o Product recommendation insights

Visualizations:

- Customer segmentation cards
- RFM analysis panels
- Product association charts
- Recommendation insights

3. FuturePrediction

Description:



Fig.no.7.2.3 Future Predictions Dashboard

This tab analyzes future business trends and demand forecasting using predictive analytics.

Key Features:

- Product demand forecasting
- Seasonal demand prediction
- Revenue scenario comparison
- Forecast confidence analysis

Visualizations:

- Demand growth bar chart
- Seasonal demand heatmap
- Revenue forecast comparison chart
- Forecast confidence gauge

4. WholesaleIntelligen

Description:

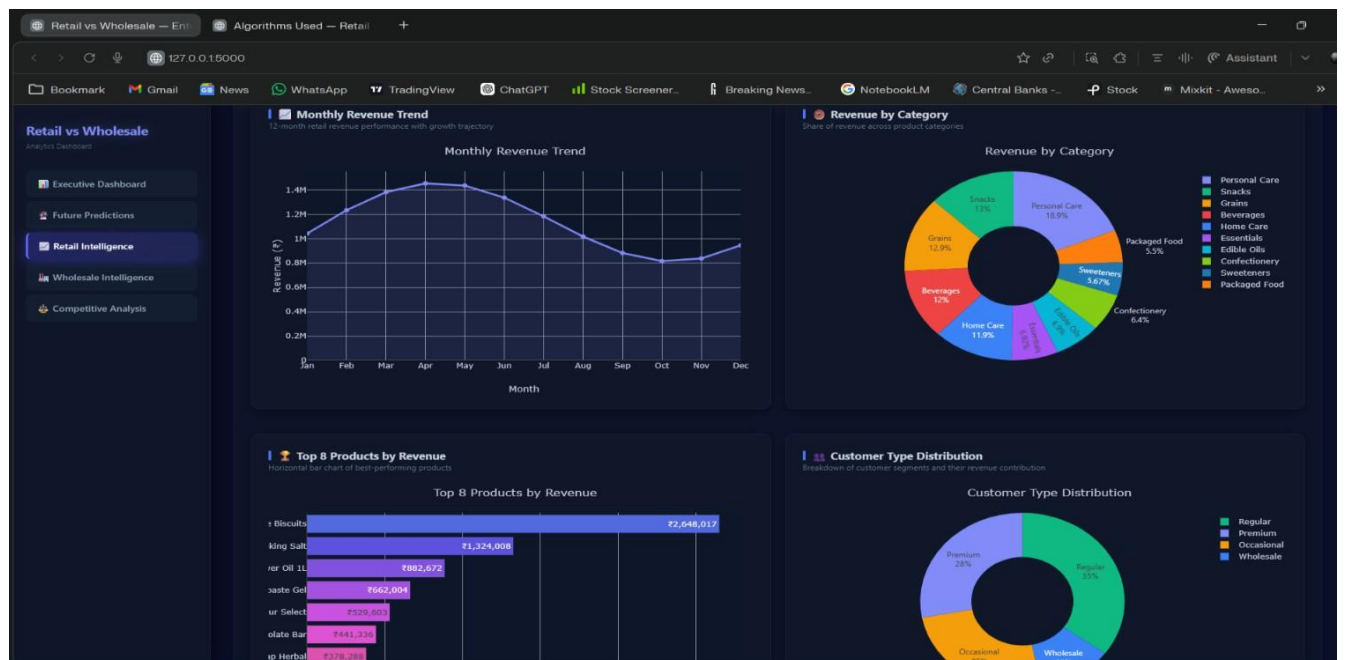


Fig.no.7.2.4 Wholesale Intelligence Dashboard

Key Features:

- Region-wise revenue distribution
- Shift-wise sales analysis
- Profit vs discount analysis
- Inventory status tracking

Visualizations:

- Revenue share bar chart
- Shift distribution pie chart
- Profit margin scatter plot
- Inventory status stacked chart

5. Competitive Analysis

Description:

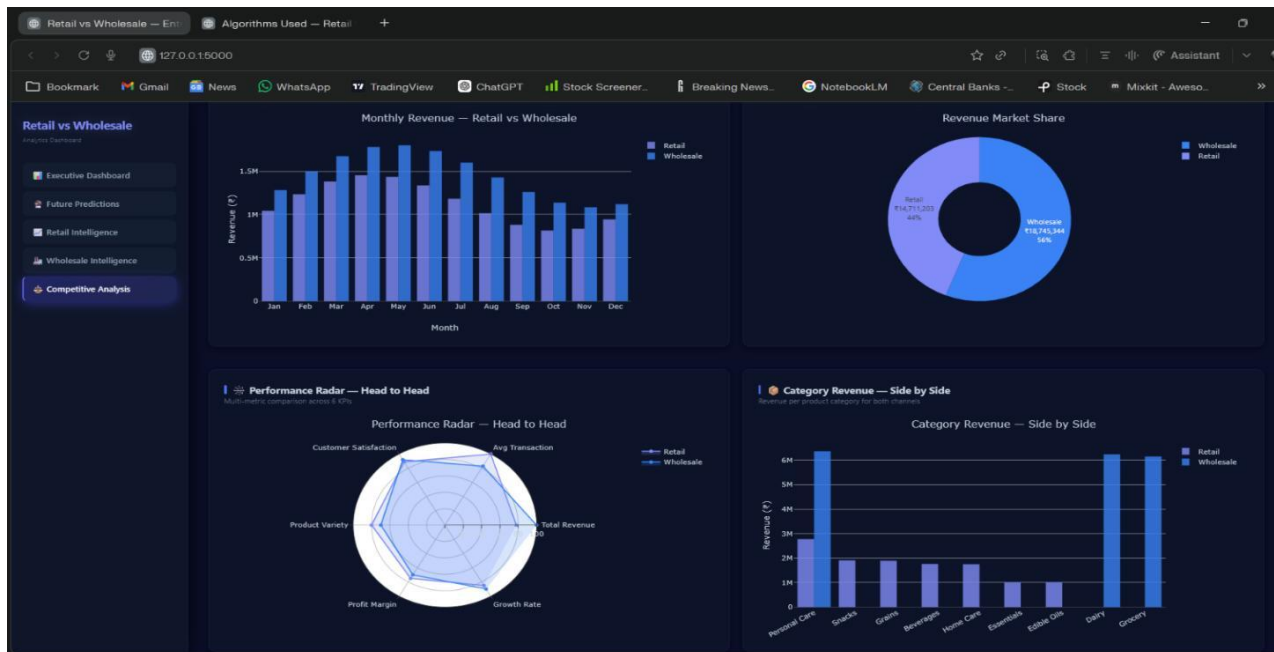


Fig.no.7.2.5 Competitive AnalysisDashboard

This tab analyzes **migration interruptions and recovery patterns**.

KEY Features:

This tab compares retail and wholesale business performance across multiple metrics.

Key Features:

- Monthly revenue comparison
- Market share analysis
- Product category comparison
- Head-to-head performance evaluation

Outputs:

- Best-performing business segment
- Revenue market share
- Growth rate comparison
- Product category performance

Visualizations:

- Monthly revenue comparison chart
- Revenue market share donut chart
- Performance radar chart
- Category revenue comparison graph

7.2.1 TOOLS USED

- Python
Used as the main programming language for implementing logic and data processing.
- Plotly
Used for creating interactive visualizations such as bar charts, pie charts, and maps.
- Dash Framework
Used to build the web-based interactive dashboard and connect frontend with backend logic.
- Dash Bootstrap Components
Used for designing responsive and visually appealing UI components.
- Pandas and NumPy
Used for handling and processing the dataset efficiently.

7.2.2 DEPLOYMENT

The developed system is deployed as a web application, allowing users to access it through a browser.

Key deployment aspects include:

- The application is executed using a Python server
- The dataset is loaded dynamically from a CSV file
- Environment variables are used to manage file paths

The application can be deployed using platforms such as:

- Gunicorn (for production server)
- Render / Railway (cloud deployment platforms)

The deployment ensures:

- Accessibility from different devices
- Real-time interaction with the system
- Efficient performance and scalability.

The final deployed system provides an intuitive interface for analyzing bird migration patterns and supports data-driven decision-making.

CHAPTER 8

CONCLUSION

8.1 SUMMARY OF FINDINGS

The Retail Analysis Platform successfully analyzes and visualizes retail sales and customer data using data mining, data warehousing, and visualization techniques. The project demonstrates how large and complex retail datasets can be transformed into meaningful business insights.

Key findings from the project include:

- Sales performance is highly influenced by pricing strategies, discounts, seasonal demand, And customer preferences
- Certain regions and store locations act as major sales hubs for multiple product categories
- Market conditions, customer behavior, and seasonal variations contribute significantly to fluctuations in sales performance
- Some product categories exhibit consistent sales trends, while others show high variability depending on customer demand and promotional activities
- Clustering techniques help in identifying similar customer purchasing behaviors and product sales patterns

The interactive dashboard enables users to explore retail data effectively, making the system useful for business analysis and decision-making.

8.2 CHALLENGES FACED

- Data Preprocessing Issues
Handling missing values, inconsistent data formats, and cleaning the dataset required careful processing
- Feature Selection Complexity
Identifying relevant attributes for different analytical questions was challenging
- Algorithm Implementation
Understanding and implementing multiple machine learning algorithms correctly required significant effort
- Clustering Evaluation

- UI Development

Integrating backend analysis with an interactive dashboard using Dash required debugging and optimization

- Performance Handling

Managing large datasets and ensuring smooth visualization without delay was challenging

8.3 FUTURE ENHANCEMENTS

The project can be further improved with the following enhancements:

- Real-time Data Integration Incorporating live sales and transaction data using APIs and cloud based retail systems
- Advanced Machine Learning Models Applying deep learning techniques for more accurate sales forecasting and customer behavior prediction
- Improved Visualization Adding more advanced and interactive dashboards, charts, and 3D visualizations for business analysis
- Mobile Application Development Creating a mobile-friendly version of the retail analysis platform
- Integration with Business Intelligence (BI) Tools Enhancing analytical capabilities using tools such as Power BI and Tableau for advanced reporting and decision-making
- Prediction System Developing a recommendation system to predict future sales trends, customer preferences, and product demand

CHAPTER 9

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CHAPTER 10

APPENDICES

Appendix A: Sample Code Snippets

A.1 Data Loading and Exploration (Python)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv('Bird_Migration_with_origin.csv')
print(df.shape)
print(df.dtypes)
print(df.describe())
```

A.2 KNN Classification (Python)

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler
X = df[features]
y = df['severity_description']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3)
scaler = MinMaxScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
print(knn.score(X_test, y_test))
```